

PD-Prune: Disaggregated Prefill-Decode Pruning for Efficient Agentic LLM Serving

Anonymous Authors

Abstract

Agentic LLM workloads repeatedly fork a committed context into k candidate continuations and retain at most one. Under prefill–decode (PD) disaggregation, each candidate is prefilled and its KV cache transferred before any decode-time pruner can reject it, so pruning decisions arrive after the cost they should avoid. We propose *PD pruning disaggregation*, which separates where a pruning decision is made, where it is executed, and how pruned KV state is managed. PD-Prune assigns each candidate one of four actions (HIT, ADMIT, PROBE, REJECT) before any kernel runs, using model-free scores and KV residency on both instances; extends uncertain candidates only if a short decoded probe passes a separate threshold; and uses a copy-on-write context tree that transfers only decoder-missing pages and retains only the committed path. On Qwen3-8B, admission in front of ESC, Speculative Rejection, and DPTS reduces computed prefill by 23% while committed-answer accuracy changes by at most two of 16 problems; on MATH-500 the probe restores the coverage lost by blanket rejection (20/32 vs. 13/32) at $0.69\times$ the wall-clock of the full fan-out; on GSM8K with Qwen3-14B the retained path needs $0.72\times$ the peak KV of prefix caching at equal accuracy.

1 Introduction

Agentic LLM applications interleave generation with branching. A coding agent proposes several candidate edits or tool calls for the same repository context, executes one, and discards the rest [36, 34, 30]; a reasoning harness expands a Tree-of-Thoughts frontier or a self-consistency pool from one partial solution [35, 31, 32]. In both cases the serving system observes the same pattern at every step: a committed context (the *trunk*) is extended by k candidate continuations (the *residuals*), and at most one of them (the *spine*) survives into the next step. Most of the candidates are discarded [8, 12,

17], so the efficiency of an agentic serving system depends on how early, and at which stage, a non-surviving candidate stops consuming resources.

Production stacks increasingly serve such workloads with prefill–decode (PD) disaggregation, which places prompt processing and token generation on separate instances and moves the KV cache between them through a connector [42, 25]. Decode-time path pruning has developed in parallel: ESC stops sampling once a window agrees [17], Speculative Rejection halts low-reward partial generations at a horizon [26], DPTS early-stops children after a mini-step [10], and STOP learns a token that predicts futility from a short prefix [4]. Combined naively, these two lines produce a structural mismatch. A pruning decision is taken on the decode instance, because every existing signal reads generated tokens, while the cost the decision is meant to avoid—prefill compute on the prefill instance and KV transfer across the connector—has already been incurred. Pruning is therefore *stage-local*: each stage can only discard work that has already reached it.

Prefix caching does not remove this mismatch. Automatic prefix caching (APC) in vLLM and RadixAttention in SGLang reuse the trunk whenever its blocks are already published and resident [14, 41]; whether k siblings issued together share the trunk depends on the cache regime and on the engine’s scheduling (§2.2). Even when the trunk is fully shared, prefix caching neither rejects a residual nor controls which pages cross the connector. Program-level schedulers and KV-retention systems for agents [23, 16, 33, 9] manage the context across tool calls, and ForkKV shares a base cache across LoRA agents [29]; none of them decides, per candidate, whether it should be prefilled, transferred, or decoded.

PD pruning disaggregation. We argue that, in disaggregated agentic serving, pruning should itself

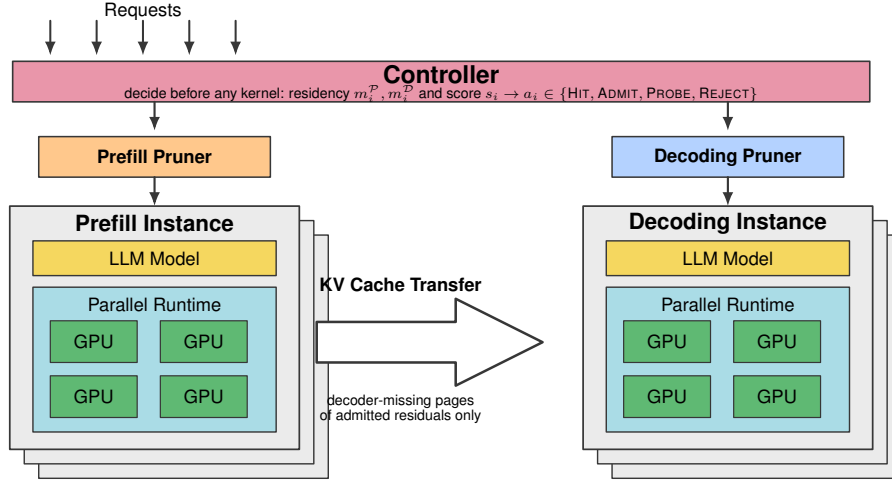


Figure 1: **PD-Prune runtime.** The controller reads the residency of each residual on the prefill instance (m_i^P) and the decode instance (m_i^D) together with a model-free score s_i , and assigns an action $a_i \in \{\text{HIT, ADMIT, PROBE, REJECT}\}$ before any kernel runs. The prefill instance computes only the non-resident suffix of admitted and probed residuals; the connector carries only the pages the decode instance lacks. The decode-side pruner extends a probed residual beyond t_0 tokens only if its generated prefix clears τ_{dec} . After commit, only the spine x_* remains resident.

be disaggregated: the *decision*, its *execution*, and the *management of pruned state* should be placed at the stages where each is cheapest and most informative. PD-Prune instantiates this principle (Figure 1).

- *Decision placement.* A controller-side admission policy assigns each residual one of four actions before any kernel runs, using only signals available at that point: the page-aligned residency of the residual on both instances and a model-free score of the residual text (§3.1).
- *Cross-stage coordination.* Residuals that cannot be resolved before prefill are *probed*: they are prefilled, decoded for t_0 tokens, and extended only if the generated prefix clears a decode-side threshold τ_{dec} that is calibrated separately from the prefill-side threshold τ_{pre} (§3.2). Decode-time pruners such as ESC, Speculative Rejection, and DPTS operate unchanged on the admitted set.
- *Pruning-aware KV management.* A copy-on-write context tree shares the trunk across siblings, transfers only decoder-missing pages, frees residual pages on abort, and retains the committed spine as the trunk of the next agent step (§B.2).

Results. On Qwen3-8B GSM8K with $k=16$, admission in front of ESC, Speculative Rejection, or DPTS reduces computed prefill from 5008 to 3848 tokens (-23%) for every host; committed-answer

accuracy changes by 0, -2 , and 0 of 16 problems, and ESC wall-clock falls from 17.05 s to 14.13 s (§4). On MATH-500, rejecting every low-scoring residual before prefill lowers any-finished coverage from 20/32 to 13/32, whereas a 128-token probe restores 20/32 at 72.2 s instead of 104.1 s. On the full GSM8K test set with Qwen3-14B, the retained spine requires $0.72\times$ the median peak KV of APC at matched accuracy, and the ratio stays within $0.72\text{--}0.75\times$ across six model families. All measurements run on physically disaggregated vLLM prefill and decode instances connected by a KV connector.

Contributions.

- We formulate pruning in disaggregated agentic serving as a per-residual, cross-stage decision problem, and make explicit the cache regimes and residency conditions under which prefix caching does or does not share the trunk (§2).
- We design PD-Prune, which disaggregates the pruning decision (pre-kernel admission), its execution (prefill, connector, and decode-side probe), and pruned-state management (copy-on-write context tree), and we give a work and transfer accounting that separates trunk sharing, admission, and probe overhead (§3).
- We show on Qwen3 models that admission composes with three decode-time pruners, that a decode-side probe recovers the coverage lost by

blanket pre-kernel rejection, and that spine retention lowers peak KV, while reporting committed-answer accuracy throughout (§4).

2 Problem Formulation

We model one agent step as a branching step served by a prefill instance $\mathcal{P}$ and a decode instance $\mathcal{D}$ connected by a KV connector. Table 7 in the appendix summarizes the notation.

Definition 1 (Branching step). A parent u holds a committed trunk x_u with $|x_u|=L$. A fan-out is a set of residuals $\{\rho_i\}_{i=0}^{k-1}$ with $|\rho_i|=\ell_i$; request i is the sequence $x_u\|\rho_i$. Index 0 denotes the residual the harness ranks first (e.g., the greedy continuation). Commit selects a winner $\star$ and leaves the spine $x_\star = x_u\|\rho_\star$, which becomes the trunk of the next step.

With b bytes of KV per token, three occupancies characterize one step:

$$\begin{aligned} M_{\text{clone}} &= b(kL + \sum_i \ell_i), \\ M_{\text{CoW}} &= b(L + \sum_i \ell_i), \\ M_{\text{spine}} &= b(L + \ell_\star). \end{aligned} \quad (1)$$

M_{clone} arises only when every request materializes its own trunk; M_{CoW} holds one shared trunk and all residuals; M_{spine} is the occupancy after the non-winners are freed.

2.1 Residency on Two Instances

Paged engines index full blocks of P tokens by a chained key $h_j = \text{hash}(h_{j-1}, \text{block}_j, \text{extra})$ [14, 41], so a block is matched only together with its entire parent prefix. For an instance $\mathcal{X} \in \{\mathcal{P}, \mathcal{D}\}$ we define the resident prefix of the trunk and of residual i as

$$\begin{aligned} c_u^{\mathcal{X}} &= \max\{jP \leq L : x_u[0:jP] \sqsubseteq \mathcal{X}\}, \\ m_i^{\mathcal{X}} &= \max\{jP \leq \ell_i : x_u\|\rho_i[0:jP] \sqsubseteq \mathcal{X}\}, \end{aligned} \quad (2)$$

where $z \sqsubseteq \mathcal{X}$ means that every page of z is physically resident on $\mathcal{X}$, and $m_i^{\mathcal{X}}=0$ whenever $c_u^{\mathcal{X}} < L$, because chained keys make a residual page unreachable without its full trunk. For exposition we take L to be a multiple of P ; otherwise the partial trunk page is charged to each residual. Three distinctions matter. (i) *Parent-aware match*: $m_i^{\mathcal{X}}$ is defined on $x_u\|\rho_i$, not on ρ_i alone. (ii) *Index presence versus residency*: an index entry whose block has been evicted does not count; $m_i^{\mathcal{X}}$ counts physically resident pages only. (iii) *Instance locality*: pages resident on $\mathcal{P}$ are not visible on $\mathcal{D}$, so $m_i^{\mathcal{P}}$ and $m_i^{\mathcal{D}}$ generally differ.

2.2 Cache Regimes

Whether the trunk is shared across the k siblings depends on the cache state and on the engine, not on prefix caching per se. Let W^{trunk} be the trunk tokens computed on $\mathcal{P}$ for one fan-out.

- R1** *Cold, concurrent*. The trunk is not resident and the siblings are admitted before any of them publishes trunk blocks; depending on how the scheduler handles in-flight blocks, $L \leq W^{\text{trunk}} \leq kL$.
- R2** *Cold, serialized*. The trunk is prefilled once (by a warm-up request or by deferring siblings until it is published); $W^{\text{trunk}}=L$ at the cost of one serialization step.
- R3** *Warm*. The trunk is resident on $\mathcal{P}$, e.g., because it is the previous spine; $W^{\text{trunk}}=0$.
- R4** *Explicit fork*. The harness declares the fan-out and the trunk is aliased copy-on-write; $W^{\text{trunk}}=L - c_u^{\mathcal{P}}$ without serialization.

APC attains R2 or R3 in many agentic deployments, so trunk sharing alone is not the source of the savings we report. All GPU measurements in §4 warm the trunk before the fan-out (R3); measured prefill reductions are therefore attributable to residual admission. On the connector, the trunk needs to cross at most once, $X^{\text{trunk}}=L - c_u^{\mathcal{D}}$, and in a multi-step agent it is typically already resident on $\mathcal{D}$ because it was decoded there.

2.3 Objective

Each residual receives an action a_i from a set $\mathcal{A}$, inducing computed prefill W_i , connector tokens X_i , and decode tokens d_i ; we write $W = \sum_i W_i$, $X = \sum_i X_i$, and $D^+ = \sum_i d_i$. A pruning policy π solves

$$\begin{aligned} \min_{\pi} \quad & \mathbb{E}[T_{\text{pre}}(W) + \mu X + T_{\text{dec}}(D^+)] \\ \text{s.t.} \quad & \mathbb{P}[\text{committed answer correct}] \geq Q_{\min}, \\ & M_{\text{peak}} \leq M_{\text{pool}}, \quad a_0 \neq \text{REJECT}, \end{aligned} \quad (3)$$

where μ is the per-token transfer time. The defining constraint is informational: a decision taken before prefill may depend only on $(c_u^{\mathcal{P}}, c_u^{\mathcal{D}}, m_i^{\mathcal{P}}, m_i^{\mathcal{D}})$ and the residual text, whereas a decision taken on $\mathcal{D}$ may additionally read generated tokens. Decode-time pruners act only at the second point; PD-Prune adds the first and coordinates the two. PD-Prune is a threshold heuristic for (3), not an optimal solver.

3 PD-Prune: PD Pruning Disaggregation

PD-Prune separates three concerns that a stage-local pruner conflates. The *decision* is taken on the controller before any kernel runs (§3.1); residuals that cannot be resolved at that point are deferred to a decode-side *probe* (§3.2); and the *state* of admitted, probed, and rejected residuals is managed by a copy-on-write context tree (§3.4). §3.3 shows how the three combine in prefill and transfer volume.

3.1 Pre-Kernel Admission

For each residual the controller observes $o_i = (m_i^{\mathcal{P}}, m_i^{\mathcal{D}}, s_i)$, where $m_i^{\mathcal{X}}$ is the parent-aware resident prefix of (2) and $s_i = \text{Score}_{\text{pre}}(\rho_i) \in [0, 1]$ is a model-free score computed from the residual text in $O(\ell_i)$ CPU time. Our scorer combines repetition detection and syntactic checks (Appendix B.4); a learned scorer can replace it without changing the policy.

Definition 2 (Admission action). Given thresholds $\tau_{\text{pre}} \leq \tau_{\text{dec}}$, residual i receives the first matching action in

$$a_i = \begin{cases} \text{HIT} & m_i^{\mathcal{D}} = \ell_i, \\ \text{ADMIT} & i = 0 \text{ or } s_i \geq \tau_{\text{dec}}, \\ \text{REJECT} & s_i < \tau_{\text{pre}}, \\ \text{PROBE} & \text{otherwise.} \end{cases} \quad (4)$$

HIT requires residency on the *decode* instance, since a page resident only on $\mathcal{P}$ must still be transferred. Because $i=0$ is caught by the first two cases, $a_0 \in \{\text{HIT}, \text{ADMIT}\}$: the top-ranked residual can be neither rejected nor probed away. Residuals in the uncertainty band $[\tau_{\text{pre}}, \tau_{\text{dec}})$ are probed. Setting $\tau_{\text{pre}} = \tau_{\text{dec}}$ empties the band and yields *blanket rejection*; replacing PROBE by ADMITYields *reject-only admission*. Let $e_i = [\text{Score}_{\text{dec}}(y_i) \geq \tau_{\text{dec}}]$ indicate that the probe y_i of a probed residual is extended (§3.2). The action determines the per-residual costs

$$W_i = \begin{cases} 0 & \text{HIT, REJECT,} \\ \ell_i - m_i^{\mathcal{P}} & \text{ADMIT,} \\ \ell_i - m_i^{\mathcal{P}} + e_i W_i^{\text{ext}} & \text{PROBE,} \end{cases} \quad (5)$$

$$X_i = (\ell_i - m_i^{\mathcal{D}}) [a_i \in \{\text{ADMIT, PROBE}\}], \quad (6)$$

$$d_i = \begin{cases} 0 & \text{REJECT,} \\ D & \text{HIT, ADMIT,} \\ t_0 + e_i(D - t_0) & \text{PROBE,} \end{cases} \quad (7)$$

Table 1: Scores of the model-free heuristic scorers on the three residual classes of the evaluation mix. $\text{Score}_{\text{pre}}$ reads the residual; $\text{Score}_{\text{dec}}$ reads a probe that has not yet produced (“illegal”) or has produced (“live”) a finished answer marker.

	loop	illegal	live
$\text{Score}_{\text{pre}}(\rho_i)$	0.02	0.22	1.00
$\text{Score}_{\text{dec}}(y_i)$	0.04	0.16	0.92

where D is the decode budget, t_0 the probe length, and W_i^{ext} the tokens recomputed when a probe is extended (§3.2). The trunk is aliased (§3.4) and contributes $W^{\text{trunk}} = L - c_u^{\mathcal{P}}$ and $X^{\text{trunk}} = L - c_u^{\mathcal{D}}$ once per fan-out. A decode-time host (ESC, Speculative Rejection, DPTS) may further shorten d_i for any admitted or extended residual.

3.2 Decode-Side Probe

A probed residual is prefilled, transferred, and decoded for t_0 tokens; the decode-side pruner then scores the generated prefix, $s'_i = \text{Score}_{\text{dec}}(y_i)$, and extends the residual to D only if $s'_i \geq \tau_{\text{dec}}$. The two thresholds read different text: τ_{pre} reads the residual before generation, and τ_{dec} reads model output. They are therefore calibrated separately. A useful pair satisfies the design condition

$$\sup_{\rho \in \mathcal{C}_{\text{rej}}} s(\rho) < \tau_{\text{pre}} \leq \inf_{\rho \notin \mathcal{C}_{\text{rej}}} s(\rho), \quad (8)$$

$$\sup_{y \in \mathcal{C}_{\text{stop}}} s'(y) < \tau_{\text{dec}},$$

where $\mathcal{C}_{\text{rej}}$ is the class of residuals that should never be prefilled (here, repetition loops) and $\mathcal{C}_{\text{stop}}$ the class of probes that should not be extended. Table 1 lists the scores of our heuristic scorers on the three residual classes of our workload; with $(\tau_{\text{pre}}, \tau_{\text{dec}}) = (0.15, 0.45)$ both conditions hold. Because the heuristics assign near-constant scores per class, this separation is specific to our workload; a learned scorer can replace them on natural agent traces (§6).

Extension cost. If the probe’s pages remain resident, extension continues decoding without recomputation and $W_i^{\text{ext}} = 0$. Our prototype issues the extension as a continuation request; when the probe’s pages have been evicted in the meantime, the engine recomputes them, so $0 \leq W_i^{\text{ext}} \leq \ell_i + t_0$. This term is negligible at $k \leq 8$ but substantial at $k=16$ in our runs (Appendix D.5), and pinning probe pages until the extension decision is an engineering fix we leave to future work.

Algorithm 1 PD-Prune for one fan-out. Phase 1 runs on the controller without any kernel; phases 2–4 execute on $\mathcal{P}$, the connector, and $\mathcal{D}$.

Input: trunk x_u ; residuals $\{\rho_i\}_{i=0}^{k-1}$; residency $c_u^{\mathcal{P}}, c_u^{\mathcal{D}}, \{m_i^{\mathcal{P}}, m_i^{\mathcal{D}}\}; \tau_{\text{pre}} \leq \tau_{\text{dec}}; t_0; D$
Output: actions $\{a_i\}$; spine x_*

- 1: {Phase 1: decide (controller)}
- 2: **for** $i = 0$ **to** $k - 1$ **do**
- 3: $s_i \leftarrow \text{Score}_{\text{pre}}(\rho_i); a_i \leftarrow (4)$
- 4: **end for**
- 5: $\mathcal{A} \leftarrow \{i : a_i \in \{\text{ADMIT}, \text{PROBE}\}\}$
- 6: {Phase 2: prefill on $\mathcal{P}$ (trunk aliased)}
- 7: prefill $x_u[c_u^{\mathcal{P}}:L]$; for $i \in \mathcal{A}$ prefill $\rho_i[m_i^{\mathcal{P}}:\ell_i]$
- 8: {Phase 3: transfer decoder-missing pages}
- 9: send $x_u[c_u^{\mathcal{D}}:L]$ and, for $i \in \mathcal{A}$, $\rho_i[m_i^{\mathcal{D}}:\ell_i]$
- 10: {Phase 4: decode on $\mathcal{D}$ }
- 11: **for** i with $a_i = \text{PROBE}$ **do**
- 12: $y_i \leftarrow \text{Dec}(x_u \parallel \rho_i, t_0); s'_i \leftarrow \text{Score}_{\text{dec}}(y_i)$
- 13: **if** $s'_i < \tau_{\text{dec}}$ **then**
- 14: free the pages of $\rho_i \parallel y_i$
- 15: **end if**
- 16: **end for**
- 17: decode HIT, ADMIT, and extended residuals to D under the host pruner
- 18: $\star \leftarrow$ commit by (11); free all non-spine residual pages

3.3 Work and Transfer Accounting

We compare PD-Prune with a baseline that serves the same fan-out without admission.

Assumption 1 (Baseline). The baseline admits every residual, computes W_B^{trunk} trunk tokens according to its cache regime (§2.2), computes $\ell_i - m_i^{\mathcal{P}}$ tokens for residual i , and transfers at least the pages each request lacks on $\mathcal{D}$, i.e., $X_B \geq (L - c_u^{\mathcal{D}}) + \sum_i (\ell_i - m_i^{\mathcal{D}})$.

Proposition 1 (Accounting). *Fix a fan-out and a cache state, and let $\mathcal{R} = \{i : a_i = \text{REJECT}\}$, $\mathcal{H} = \{i : a_i = \text{HIT}\}$, and $\mathcal{E} = \{i : a_i = \text{PROBE}, e_i = 1\}$. Under Assumption 1,*

$$W_B - W = \Delta_{\text{trunk}} + \Delta_{\text{adm}} - \Delta_{\text{ext}}, \quad (9)$$

$$X_B - X \geq \sum_{i \in \mathcal{R}} (\ell_i - m_i^{\mathcal{D}}),$$

with trunk-sharing term $\Delta_{\text{trunk}} = W_B^{\text{trunk}} - (L - c_u^{\mathcal{P}})$, admission term $\Delta_{\text{adm}} = \sum_{i \in \mathcal{R} \cup \mathcal{H}} (\ell_i - m_i^{\mathcal{P}})$, and probe overhead $\Delta_{\text{ext}} = \sum_{i \in \mathcal{E}} W_i^{\text{ext}}$.

Proof. Sum (5) and (6) over i and add the trunk terms. The baseline charges $\ell_i - m_i^{\mathcal{P}}$ for every i , while PD-Prune charges it only for $i \in \mathcal{A}$ and adds W_i^{ext} for $i \in \mathcal{E}$; the residuals not in $\mathcal{A}$ are exactly $\mathcal{R} \cup \mathcal{H}$. For transfer, HIT residuals satisfy $\ell_i - m_i^{\mathcal{D}} = 0$, so the baseline lower bound exceeds PD-Prune’s transfer by the rejected residuals’ decoder-missing pages. $\square$

The decomposition makes explicit which savings depend on the cache regime. Δ_{trunk} is zero in regimes R2–R4 and positive only when concurrent siblings recompute the trunk (R1); Δ_{adm} does not depend on the regime; Δ_{ext} is a cost. Decode-time hosts change only D^+ : since they act on generated tokens of residuals that have already been prefilled, W and X do not depend on the host. The wall-clock of a fan-out is then

$$T_{\text{fan}} = T_{\text{pre}}(W) + \mu X + T_{\text{dec}}(D^+) + T_{\text{CoW}} + T_{\text{abort}}, \quad (10)$$

where T_{CoW} and T_{abort} are page-table operations independent of L .

3.4 Pruning-Aware KV State

PD-Prune keeps one copy-on-write context tree per session (Appendix B.2, Figure 2). fork creates a child whose page table aliases the parent’s pages and increments their reference counts, an $O(1)$ operation in L ; only residual pages are allocated. A rejected residual allocates nothing; a probe that is not extended frees its residual and probe pages; abort frees the residual pages of every non-winner, bl_i bytes each. On the next observation x , commit binds the child with the longest common prefix,

$$v^* = \arg \max_v \text{LCP}(x, x_v), \quad g_u \leftarrow g_u + 1, \quad (11)$$

publishes the full pages of x_* to the prefix index, and invalidates in-flight work tagged with an older generation g_u . Speculatively prefilled pages are never sampled from, so the emitted tokens equal those of a baseline that prefills committed tokens only. After commit the live occupancy is M_{spine} , and a pool of M_{pool} bytes holds $\lfloor M_{\text{pool}}/M_{\text{spine}} \rfloor$ sessions instead of $\lfloor M_{\text{pool}}/M_{\text{APC}} \rfloor$ while a prefix cache still retains aborted residuals.

Cross-step coordination. The spine of step t is the trunk of step $t+1$. Because it was decoded on $\mathcal{D}$ and published on $\mathcal{P}$, the next fan-out starts in regime R3 or R4 with $c_u^{\mathcal{D}} = L$, so the trunk crosses the connector at most once per session rather than once per fan-out; a residual repeated across steps (e.g., a re-issued tool call) becomes a HIT. Admission is re-evaluated at every step, so rejected branches never accumulate state across turns.

4 Evaluation

Setup. All GPU runs use vLLM on A100-SXM4 40GB GPUs with bfloat16 weights, temperature

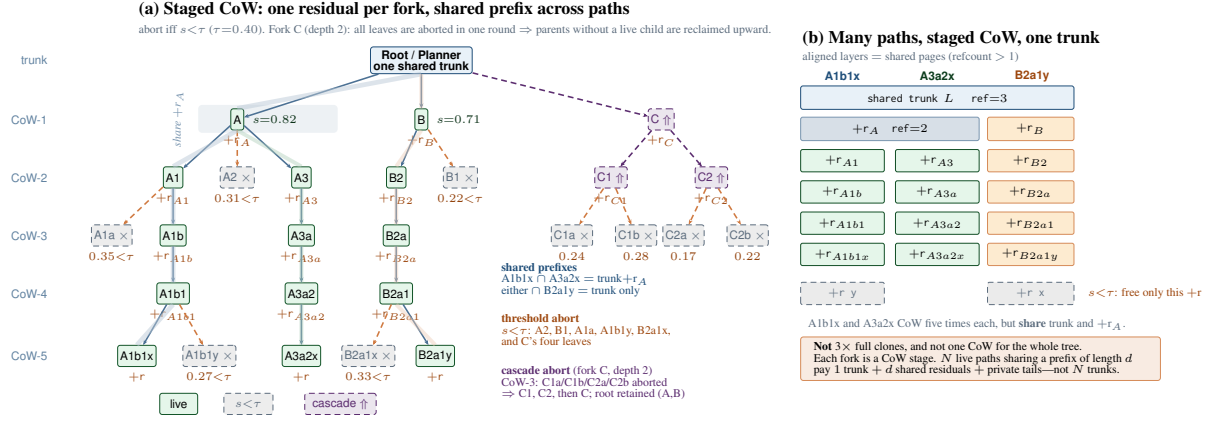


Figure 2: **Context tree.** (a) Each edge is one fork: the child aliases the parent and allocates only a residual (+r). Live paths share the trunk; A1b1x and A3a2x also share +r_A. All descendants of fork C are aborted at depth 2, so reclamation propagates up to C and stops at the root, which retains the live children A and B. (b) A shared layer is one physical page set with reference count greater than one; aborting a leaf frees only its residual.

0, and prefix caching enabled. Prefill and decode are physically disaggregated: $\mathcal{P}$ and $\mathcal{D}_{\text{run}}$ as separate vLLM instances connected by a KV connector, and admitted residuals reach $\mathcal{D}_{\text{only}}$ through that connector. We measure computed prefill W , decode tokens D^+ , peak KV, and end-to-end wall-clock. The trunk is warmed before every fan-out (regime R3), so prefill differences reflect residual admission only. The admission experiments use Qwen3-8B at $\text{tp}=1$ on a *branching mix*: for each problem, half of the k residuals are distinct solution strategies, a quarter are repetition loops, and a quarter are syntactically malformed (“illegal”) text. GSM8K uses $n=16$ problems and $D=512$; MATH-500 uses the first 32 problems of level ≥ 4 and $D=1024$. Decoding ignores EOS with a fixed budget, so D^+ is determined by the policy. Our primary metric is *committed-answer accuracy*: the answer of the residual the harness commits (index 0 for MATH-500, the host’s selection for GSM8K) matches the reference. We also report *any-finished coverage*, i.e., whether some full-budget branch is correct, which upper-bounds what a better selector could commit. With $n=16$, two runs of the identical ESC configuration at $k=16$ commit 11/16 and 8/16 (Appendix D.2), so differences of up to three problems are within run-to-run variation.

Admission composes with decode-time pruners. Table 2 (GSM8K block) places one admission policy in front of three decode-time hosts. Without admission, ESC, Speculative Rejection, and DPTS prefill the same 5008 tokens; they differ only in D^+ (131072, 98304, and 78336 tokens) because each reads generated text. Reject-only admission at

$\tau_{\text{pre}}=0.15$ rejects the four loops before any kernel and reduces W to 3848 tokens (−23%) for every host, as Proposition 1 predicts. Committed accuracy is unchanged for ESC and DPTS and two problems lower for Speculative Rejection, within the run-to-run variation stated above; coverage remains 16/16. The time reduction is largest for ESC (−17%), which would otherwise decode the loops to the full budget, and small for hosts that already stop them early (−5%, −3%). Blanket rejection at 0.45 also removes the illegal residuals and is faster (−36% for ESC) at a cost of one to two committed problems at $k=16$. Table 3 shows that the trade-off depends on width: at $k=4$ and $k=8$ blanket rejection loses two problems of coverage, while reject-only stays within one.

The probe recovers what blanket rejection loses. On MATH-500 (Table 2, lower block), blanket rejection halves the time of the full fan-out but lowers coverage from 20/32 to 13/32: some residuals scored as malformed nevertheless lead to correct solutions once decoded. Reject-only admission recovers part of the loss (16/32). Probing the uncertainty band with $t_0=128$ restores 20/32 at 72.2 s, $0.69\times$ the full fan-out, and reduces D^+ from 524288 to 393216 tokens. All 128 probed residuals were extended, and no loop residual would have passed the extension threshold (Appendix D.5), so on this mix the probe acts as a correction of the prefill-side score rather than as an additional filter. Its cost is visible in W : at $k=16$, evicted probe pages are recomputed on extension and W rises to 28736 tokens, which (9) accounts for as Δ_{ext} ; at $k \leq 8$ the overhead is below 300 tokens. The best probe

Table 2: Main results on Qwen3-8B (branching mix, trunk warm). *GSM8K* ($k=16, n=16, D=512$): “Reject-only” rejects $s_i < 0.15$ and admits the rest; “Blanket” rejects $s_i < 0.45$. Prefill W is identical across hosts because hosts act after prefill. *MATH-500* ($k=16, n=32, D=1024$, no host): “Probe” uses $(\tau_{\text{pre}}, \tau_{\text{dec}}, t_0) = (0.15, 0.45, 128)$; its W includes W^{ext} from evicted probe pages. The *MATH-500* committed residual is fixed to index 0, so committed accuracy is insensitive to admission by construction and coverage is the informative metric.

Policy	admitted	W	time (s)	committed	any-finished
<i>GSM8K; columns list ESC / Speculative Rejection / DPTS</i>					
host only	16	5008	17.05 / 13.14 / 11.86	11 / 12 / 12	16 / 16 / 16
+ Reject-only	12	3848	14.13 / 12.47 / 11.49	11 / 10 / 12	16 / 16 / 16
+ Blanket	8	2768	10.92 / 10.80 / 10.90	10 / 10 / 10	16 / 16 / 16
<i>MATH-500; out of 32</i>					
full fan-out	16	12320	104.1	3	20
Blanket	8	6200	52.1	3	13
Reject-only	12	9332	71.4	2	16
Probe	8+4	28736	72.2	3	20

Table 3: ESC on GSM8K across fan-out width (Qwen3-8B, $n=16$). Each cell: committed / any-finished / seconds.

	$k=4$	$k=8$	$k=16$
ESC	9 / 16 / 8.65	11 / 16 / 10.84	11 / 16 / 17.05
+ Reject-only (0.15)	9 / 15 / 8.22	9 / 16 / 10.19	11 / 16 / 14.13
+ Blanket (0.45)	11 / 14 / 7.89	9 / 14 / 8.60	10 / 16 / 10.92

Table 4: *MATH-500* coverage (any-finished out of 32) and seconds across width. “Blanket” rejects $s_i < 0.45$; “Reject-only” rejects loops; “Probe t_0 ” probes the band $[0.15, 0.45]$.

	$k=4$	$k=8$	$k=16$
Full fan-out	12, 25.3	15, 52.5	20, 104.1
Blanket	6, 19.1	11, 25.6	13, 52.1
Reject-only	11, 23.0	17, 41.5	16, 71.4
Probe $t_0=64$	8, 24.0	16, 42.0	18, 72.5
Probe $t_0=128$	9, 24.7	13, 43.3	20, 72.2

length depends on width (Table 4): $t_0=64$ is preferable at $k=8$, and at $k=4$ no probe matches the full fan-out.

Connector volume. Table 5 evaluates Proposition 1 for a fan-out with $k=4, L=256, \ell_i=32$, and one rejected loop. In a cold regime the dominant term is trunk sharing, whose size depends on whether the baseline engine deduplicates concurrent siblings and whether its connector is residency-aware. In the warm regime that characterizes multi-step agents, the trunk is already on $\mathcal{D}$ and the only remaining saving is admission, here 25% of the residual volume. We therefore attribute connector savings to admission only in the warm column and treat the cold column as an upper bound that a deduplicating, residency-aware baseline largely closes.

Spine retention lowers peak KV. On the full GSM8K test set ($n=1319$, Qwen3-14B, $\text{tp}=2$,

Table 5: Prefill and connector tokens per fan-out from Proposition 1 ($k=4, L=256, \ell_i=32$, one residual rejected). X_B is given for a residency-aware connector (lower bound) and one that ships each request in full.

Regime	W_B	W	X_B (aware / full)	X
Cold (R1-R2)	384-1152	352	384 / 1152	352
Warm (R3)	128	96	128 / 1152	96

$k=4$), the context tree commits the same answers as APC within one point (0.845 vs. 0.844) at $0.72 \times$ its median peak KV (1030 vs. 1430 tokens; Appendix C.1). This result is due to state management alone (CoW-only, i.e., PD-Prune without admission), since admission does not change the committed spine. The ratio follows $(L + \ell_*) / (L + \sum_i \ell_i)$ and is therefore governed by k and the trunk-to-residual ratio rather than by the model: it is $0.72-0.75 \times$ on GSM8K and $0.57-0.61 \times$ on Game of 24 across six families (Figure 12 in Appendix C.8), and it falls from about 0.90 at $k=2$ to $0.53-0.55$ at $k=8$ (Appendix C.9). The copy-on-write fan-out incurs its own overhead: its median latency is 140 ms, compared with 87 ms for an APC hash hit.

Multi-step agent sessions. We run a three-step Tree-of-Thoughts session ($k=4$ per step, $n=16$, Qwen3-4B and Qwen3-8B, $\text{tp}=2$) in which each step’s spine is the next step’s trunk (Table 6). Admission runs before every fan-out and prefills 48 of the 192 residuals. Peak KV is $0.86 \times$ APC on GSM8K and $0.82 \times$ on Game of 24 for both CoW-only and PD-Prune. Committed accuracy matches APC on Qwen3-4B GSM8K, is one problem lower on Qwen3-8B GSM8K (15 vs. 16 of 16), and is one to three problems higher on Game of 24. The PD-Prune runs additionally stop committed decode at a finished answer marker, so their end-to-end reductions ($0.70-0.77 \times$ APC on three of four set-

Table 6: Three-step Tree-of-Thoughts ($n=16$, $k=4$, $tp=2$). Accuracy is committed-answer accuracy; e2e is seconds per problem; peak is spine tokens. PD-Prune runs include an answer-marker decode stop.

	Qwen3-4B			Qwen3-8B		
	APC	CoW-only	PD-Prune	APC	CoW-only	PD-Prune
GSM8K acc.	0.938	0.938	0.938	1.000	0.938	0.938
GSM8K e2e	1.66	1.76	1.63	2.48	2.56	1.91
GSM8K peak	318	275	275	318	275	275
Game24 acc.	0.375	0.500	0.562	0.562	0.562	0.625
Game24 e2e	1.61	1.64	1.13	2.15	2.15	1.52
Game24 peak	265	216	216	265	216	216

tings) combine admission with decode truncation and should not be attributed to admission alone.

5 Related Work

Inference-time search and path pruning. Tree-of-Thoughts, self-consistency, and Monte Carlo tree search trade tokens for accuracy by expanding many partial solutions [35, 31, 39, 32]. ESC, Speculative Rejection, and DPTS stop branches after an agreeing window, a reward horizon, or a mini-step [17, 26, 10]; STOP organizes such signals by source and learnability and trains an internal futility token [4]; EquivPruner merges semantically equivalent steps [21]. All of these decide from generated text and are agnostic to where the KV cache was computed. PD-Prune is complementary: it adds a pre-kernel decision point and a probe that hosts these pruners unchanged, and a learned scorer such as STOP’s could serve as $\text{Score}_{\text{dec}}$.

Prefix sharing and disaggregation. vLLM and SGLang share prefixes through block hashing and radix trees [14, 41], Orca and Sarathi schedule at iteration and chunk granularity [37, 2], and DistServe and Splitwise separate prefill from decode [42, 25]. These mechanisms determine whether a trunk is shared and how KV moves, but they serve every request they receive; PD-Prune decides which residuals become requests and which pages need to move, using residency on both instances.

Agentic serving. Agentix and Hermes schedule at program granularity [23, 22]; ThunderAgent and SAGA optimize workflows [13, 11]; InferCept, Continuum, and MORI retain KV across tool calls [1, 16, 33]; ConServe schedules disaggregated prefill and decode at conversation granularity from observed agent behavior [9]; Sutradhara overlaps tool execution with the prefill of a single tool-independent prefix [5]. ForkKV is closest in mech-

anism: it shares a base KV cache copy-on-write across LoRA agents [29]. These systems place, schedule, or retain contexts that will be served; ConServe in particular disaggregates scheduling but not pruning. PD-Prune targets the complementary decision of which sibling continuations should be served at all, and at which PD stage that decision is made. Appendix E extends this discussion.

6 Conclusion

In disaggregated agentic serving, the cost of a discarded continuation is incurred at the prefill instance and the connector, whereas existing pruning signals are evaluated at the decode instance. PD-Prune disaggregates pruning accordingly: it decides before any kernel from residency and model-free scores, defers uncertain candidates to a decode-side probe with a separately calibrated threshold, and manages the resulting KV state with a copy-on-write context tree that retains only the committed spine. The accounting in Proposition 1 separates the savings that depend on the cache regime from those due to admission. Empirically, admission reduces computed prefill by 23% in front of three decode-time pruners within the run-to-run variation of committed accuracy, the probe recovers the coverage lost by blanket rejection, and spine retention lowers peak KV to $0.72\times$ that of prefix caching.

Limitations

Our scorers are lightweight heuristics evaluated on a controlled branching mix; natural agent residuals may be less separable, and learned scorers such as STOP [4] can be substituted without changing the admission policy. The admission experiments use moderate sample sizes ($n=16$ and $n=32$), so small differences in committed accuracy fall within run-to-run variation. Extending the evaluation from reasoning and Tree-of-Thoughts sessions to tool-using agents with real environments is a natural next step.

References

- [1] Reyna Abhyankar, Zijian He, Vikranth Srivatsa, Hao Zhang, and Yiyang Zhang. InferCept: Efficient Inter-cept Support for Augmented Large Language Model Inference. In *ICML*, 2024.
- [2] Amey Agrawal, Nitin Kedia, Ashish Panwar, Jayashree Mohan, Nipun Kwatra, Bhargav Gulavani,

- Alexey Tumanov, and Ramachandran Ramjee. Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve. In *OSDI*, 2024.
- [3] Amey Agrawal, Ashish Panwar, Jayashree Mohan, Nipun Kwatra, Bhargav S. Gulavani, and Ramachandran Ramjee. SARATHI: Efficient LLM Inference by Piggybacking Decodes with Chunked Prefills. *arXiv:2308.16369*, 2023.
- [4] Jiayi Bi, Tongxu Luo, Wenyu Du, Zhengyang Tang, and Benyou Wang. Cut Your Losses! Learning to Prune Paths Early for Efficient Parallel Reasoning. In *ACL*, 2026.
- [5] Anish Biswas, Kanishk Goel, Srivarshinee S, Jayashree Mohan, Alind Khare, Anjali Parayil, Ramachandran Ramjee, and Chetan Bansal. Sutradhara: An Intelligent Orchestrator-Engine Co-design for Tool-based Agentic Inference. *arXiv:2601.12967*, 2026.
- [6] Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple LLM Inference Acceleration Framework with Multiple Decoding Heads. In *ICML*, 2024.
- [7] Daewon Choi, Kyunghyun Park, Woomin Song, Saket Dingliwal, Sai Muralidhar Jayanthi, Jinwoo Shin, and Aram Galstyan. IdleSpec: Exploiting Idle Time via Speculative Planning for LLM Agents. *arXiv:2605.22154*, 2026.
- [8] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training Verifiers to Solve Math Word Problems. *arXiv:2110.14168*, 2021.
- [9] Jianru Ding, Ryien Hosseini, Pouya Mahdi Gholami, Mingyuan Xiang, and Henry Hoffmann. Observation, Not Prediction: Conversation-Level Disaggregated Scheduling for Agentic Serving. *arXiv:2606.01839*, 2026.
- [10] Yifu Ding, Wentao Jiang, Shunyu Liu, Yongcheng Jing, Jinyang Guo, Yingjie Wang, Jing Zhang, Zeng-mao Wang, Ziwei Liu, Bo Du, Xianglong Liu, and Dacheng Tao. Dynamic Parallel Tree Search for Efficient LLM Reasoning. In *ACL*, 2025.
- [11] Dongxin Guo, Jikun Wu, and Siu-Ming Yiu. SAGA: Workflow-Atomic Scheduling for AI Agent Inference on GPU Clusters. In *HPDC*, 2026.
- [12] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring Mathematical Problem Solving with the MATH Dataset. In *NeurIPS Datasets and Benchmarks*, 2021.
- [13] Hao Kang, Ziyang Li, Xinyu Yang, Weili Xu, Yinfang Chen, Junxiong Wang, Beidi Chen, Tushar Krishna, Chenfeng Xu, and Simran Arora. ThunderAgent: A Fast, Simple, and Program-Aware Agentic Inference System. In *ICML*, 2026.
- [14] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *SOSP*, 2023.
- [15] Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast Inference from Transformers via Speculative Decoding. In *ICML*, 2023.
- [16] Hanchen Li, Runyuan He, Qiuyang Mang, Qizheng Zhang, Huanzhi Mao, Xiaokun Chen, Hangrui Zhou, Huanchen Zhang, Alvin Cheung, Joseph Gonzalez, and Ion Stoica. Continuum: Efficient and Robust Multi-Turn LLM Agent Scheduling with KV Cache Time-to-Live. *arXiv:2511.02230*, 2025.
- [17] Yiwei Li, Peiwen Yuan, Shaoxiong Feng, Boyuan Pan, Xinglin Wang, Bin Sun, Heda Wang, and Kan Li. Escape Sky-High Cost: Early-Stopping Self-Consistency for Multi-Step Reasoning. In *ICLR*, 2024.
- [18] Yuhong Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. EAGLE: Speculative Sampling Requires Rethinking Feature Uncertainty. In *ICML*, 2024.
- [19] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s Verify Step by Step. In *ICLR*, 2024.
- [20] Chaofan Lin, Zhenhua Han, Chengruidong Zhang, Yuqing Yang, Fan Yang, Chen Chen, and Lili Huang. Parrot: Efficient Serving of LLM-based Applications with Semantic Variable. In *OSDI*, 2024.
- [21] Jiawei Liu, Qisi Chen, Jianshu Zhang, Quan Liu, and Defu Lian. EquivPruner: Boosting Efficiency and Quality in LLM-Based Search via Action Pruning. In *ACL*, 2026.
- [22] Yifei Liu, Zuo Gan, Zhenghao Gan, Weiye Wang, Chen Chen, Yizhou Shan, Xusheng Chen, Zhenhua Han, Yifei Zhu, Shixuan Sun, and Minyi Guo. Hermes: Efficient Serving of LLM Applications with Probabilistic Demand Modeling. *ACM Transactions on Architecture and Code Optimization*, 23(2), 2026.
- [23] Michael Luo, Xiaoxiang Shi, Colin Cai, Tianjun Zhang, Justin Wong, Yichuan Wang, Chi Wang, Yanping Huang, Zhifeng Chen, Joseph E. Gonzalez, and Ion Stoica. Agentix: An Efficient Serving Engine for LLM Agents as General Programs. In *NSDI*, 2026.
- [24] Zaifeng Pan, Qianxu Wang, Zhengding Hu, Chang Chen, Yue Guan, Yanbo Zhou, Steven Swanson, and Yufei Ding. SmoothAgent: Efficient Long-Horizon LLM-Based Agent Serving with Lookahead Context Engineering. *arXiv:2607.00151*, 2026.
- [25] Pratyush Patel, Esha Choukse, Chaojie Zhang, Aashaka Shah, Íñigo Goiri, Saeed Maleki, and Riccardo Bianchini. Splitwise: Efficient Generative LLM Inference Using Phase Splitting. In *ISCA*, 2024.

- [26] Hanshi Sun, Momin Haider, Ruiqi Zhang, Huitao Yang, Jiahao Qiu, Ming Yin, Mengdi Wang, Peter Bartlett, and Andrea Zanette. Fast Best-of-N Decoding via Speculative Rejection. In *NeurIPS*, 2024.
- [27] Xin Tan, Yimin Jiang, Yitao Yang, and Hong Xu. Towards End-to-End Optimization of LLM-based Applications with Ayo. In *ASPLOS*, 2025.
- [28] Rui Wang, Junda Wu, Yu Xia, Tong Yu, Ruiyi Zhang, Ryan Rossi, Subrata Mitra, Lina Yao, and Julian McAuley. CachePrune: Teaching LLMs What Not to Follow via KV-Cache Editing. In *ACL*, 2026.
- [29] Shao Wang, Rui Ren, and Lin Gui. ForkKV: Scaling Multi-LoRA Agent Serving via Copy-on-Write Disaggregated KV Cache. *arXiv:2604.06370*, 2026.
- [30] Xingyao Wang, Boxuan Li, Yufan Song, Frank F. Xu, Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi Song, Bowen Li, Jaskirat Singh, Hoang H. Tran, Fuqiang Li, Ren Ma, Mingzhang Zheng, Bill Qian, Yanjun Shao, Niklas Muennighoff, Yizhe Zhang, Binyuan Hui, Junyang Lin, Robert Brennan, Hao Peng, Heng Ji, and Graham Neubig. OpenHands: An Open Platform for AI Software Developers as Generalist Agents. In *ICLR*, 2025.
- [31] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-Consistency Improves Chain of Thought Reasoning in Language Models. In *ICLR*, 2023.
- [32] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *NeurIPS*, 2022.
- [33] Tian Xia, Hanchen Li, Zhifei Li, Xiaokun Chen, Hao Kang, Yifan Qiao, Yi Xu, and Ion Stoica. Idleness is Relative: Exploiting Tool-Call Idle Windows for Offloading in Agentic Systems with MORI. *arXiv:2606.00866*, 2026.
- [34] John Yang, Carlos E. Jimenez, Alexander Wetig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. In *NeurIPS*, 2024.
- [35] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. In *NeurIPS*, 2023.
- [36] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing Reasoning and Acting in Language Models. In *ICLR*, 2023.
- [37] Gyeong-In Yu, Joo Seong Jeong, Geon-Woo Kim, Soojeong Kim, and Byung-Gon Chun. Orca: A Distributed Serving System for Transformer-Based Generative Models. In *OSDI*, 2022.
- [38] Shan Yu, Junyi Shu, Yuanjiang Ni, Kun Qian, Xue Li, Yang Wang, Jinyuan Zhang, Ziyi Xu, Shuo Yang, Lingjun Zhu, Ennan Zhai, Qingda Lu, Jiarong Xing, Youyou Lu, Xin Jin, Xuanzhe Liu, and Harry Xu. Pythia: Exploiting Workflow Predictability for Efficient Agent-Native LLM Serving. *arXiv:2604.25899*, 2026.
- [39] Di Zhang, Xiaoshui Huang, Dongzhan Zhou, Yuqiang Li, and Wanli Ouyang. Accessing GPT-4 Level Mathematical Olympiad Solutions via Monte Carlo Tree Self-refine with LLaMa-3 8B. *arXiv:2406.07394*, 2024.
- [40] Wei Zhang, Zhiyu Wu, Yi Mu, Rui Ning, Banruo Liu, Nikhil Sarda, Myungjin Lee, and Fan Lai. JIT-Serve: SLO-aware LLM Serving with Imprecise Request Information. In *NSDI*, 2026.
- [41] Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. SGLang: Efficient Execution of Structured Language Model Programs. In *NeurIPS*, 2024.
- [42] Yinmin Zhong, Shengyu Liu, Junda Chen, Jianbo Hu, Yibo Zhu, Xuanzhe Liu, Xin Jin, and Hao Zhang. DistServe: Disaggregating Prefill and Decoding for Goodput-optimized Large Language Model Serving. In *OSDI*, 2024.

A Background

A.1 Branching in Agentic and Reasoning Workloads

Chain-of-thought produces one trace [32]; Tree-of-Thoughts, self-consistency, and Monte Carlo tree search produce many and then select [35, 31, 39]. In the Game-of-24 setting of Tree-of-Thoughts, a proposal prompt emits candidate equations, a value prompt labels each candidate, and breadth-first search retains a beam [35]. Agent harnesses follow the same pattern: a coding or tool-using agent proposes several continuations of one context, executes or observes one, and discards the rest [36, 34, 30]. On GSM8K and MATH the accuracy gain of such width is well documented [8, 12, 19], and so is its token cost [21, 4].

A.2 Decode-Time Path Pruning

Early-stopping self-consistency draws a window of samples and stops once the window agrees [17, 31]. Speculative Rejection ranks partial generations at a reward horizon and halts the lower quantile [26]. DPTS generates a mini-step on every active node and early-stops low-confidence children [10]. STOP organizes these signals by source and learnability and trains an internal token that

predicts futility from a short prefix [4]; Equiv-Pruner merges semantically equivalent actions [21]. Each signal is a function of generated text. On our GSM8K mix at $k=16$, ESC, Speculative Rejection, and DPTS therefore prefill the same 5008 tokens and differ only in decode volume (131072, 98304, and 78336 tokens).

A.3 Prefix Caching and Disaggregated Transfer

APC and RadixAttention index full blocks by chained keys, $h_j = \text{hash}(h_{j-1}, \text{block}_j, \text{extra})$ [14, 41]. A request reuses a block only if the block and its entire parent chain are resident. Whether k siblings issued together share the trunk depends on the cache regime of §2.2: a resident trunk (R3) is shared by all siblings; a cold trunk is computed once if siblings are serialized behind it (R2), and between one and k times if they are admitted concurrently, depending on how the scheduler treats blocks that are still being computed (R1). Prefix caching thus addresses trunk reuse, but it neither rejects a residual nor decides which pages cross a connector.

PD disaggregation places prefill and decode on separate instances and transfers KV between them [42, 25]. Every admitted request is prefilled on $\mathcal{P}$, and the pages that $\mathcal{D}$ lacks are transferred; a residual that the harness later discards has already consumed both. Program- and workflow-level schedulers raise the scheduling unit above the request [23, 13, 16, 33], and KV retention removes trunk recomputation across tool gaps; the remaining per-step cost is the residual suffixes of siblings that the harness can enumerate before the observation returns.

A.4 Prefill, Decode, and Idle Time

Prefill processes the prompt in parallel and is compute-bound; decode emits one token per step and is memory-bandwidth-bound [37, 42, 25]. PagedAttention stores KV in non-contiguous pages [14], and chunked prefill co-schedules a prompt chunk with a decode batch [2]. For agents, the prompt of step $t+1$ extends that of step t after a tool interval of variable length [16, 33], during which known suffixes can be prefilled (Appendix B.5). Figure 3 shows where each existing mechanism can first discard a sibling, and Table 7 summarizes the notation.

Table 7: Notation.

Symbol	Domain	Meaning
L, ℓ_i, ℓ_*	$\mathbb{N}$	trunk; residual i ; committed residual
x_u, ρ_i, x_*	tokens	trunk; residual; spine
$\mathcal{P}, \mathcal{D}$	—	prefill and decode instances
c_u^x, m_i^x	tokens	resident prefix of trunk and residual on $\mathcal{X}$
s_i, s'_i	$[0, 1]$	residual score; probe score
$\tau_{\text{pre}}, \tau_{\text{dec}}$	$[0, 1]$	rejection and extension thresholds
a_i	$\mathcal{A}$	HIT, ADMIT, PROBE, REJECT(Definition 2)
e_i	$\{0, 1\}$	probe of i is extended
t_0, D, d_i	tokens	probe length; budget; decode tokens of i
W_i, X_i	tokens	computed prefill; connector tokens
W_i^{ext}	tokens	recomputation on probe extension
$\mathcal{R}, \mathcal{H}, \mathcal{E}$	$\subseteq [k]$	rejected, hit, and extended residuals
b, P	bytes, tokens	KV bytes per token; page size
M_{pool}	bytes	KV pool capacity
$B_t = B_t^c + B_t^s$	tokens	committed and speculative chunk budget

B System Details

PD-Prune sits between the agent harness and the LLM engine (Figure 1). The harness exposes branch structure, the controller decides and binds, and the engine executes prefill, transfer, and decode.

B.1 Interface

open(session)	Create a root; prefill the initial context.
fork(u , bid, tok?)	Child of u , optionally with a known suffix.
speculate(v , pri)	Mark v eligible for idle-time prefill.
commit(u , tokens)	Bind by LCP ((11)); start decode.
join(children, pol)	Merge a fan-out (all/first/kofn).
abort(v) / close	Drop a subtree; end the session.

Only commit lies on the critical path of user-visible tokens.

B.2 Copy-on-Write Context Tree

Definition 3 (Context tree). A session is a rooted tree $T = (V, E)$. Node u stores a token sequence x_u with $x_{\pi(u)} \preceq x_u$, a residual $\delta_u = x_u[|x_{\pi(u)}|:]$, a mode in $\{\text{SPEC}, \text{COMMIT}, \text{IDLE}, \text{DEAD}\}$, a generation g_u , and a page table $\sigma_u = (\text{parent}, \text{alias_len}, \text{pages}_u)$.

The tree maintains four invariants: (i) prefix closure; (ii) at most one committed child per node outside a join; (iii) speculative nodes are never sampled from; (iv) the reference count of a page equals the number of non-dead nodes that map it. $\text{fork}(u, \cdot)$ writes two words, ($\text{parent} \leftarrow \sigma_u$, $\text{alias_len} \leftarrow |x_u|$), and increments the reference counts of the aliased pages; a write to a shared or read-only page copies only its valid rows. A multi-level fan-out is a sequence of such forks (Figure 2); abort frees residual pages only, and a subtree

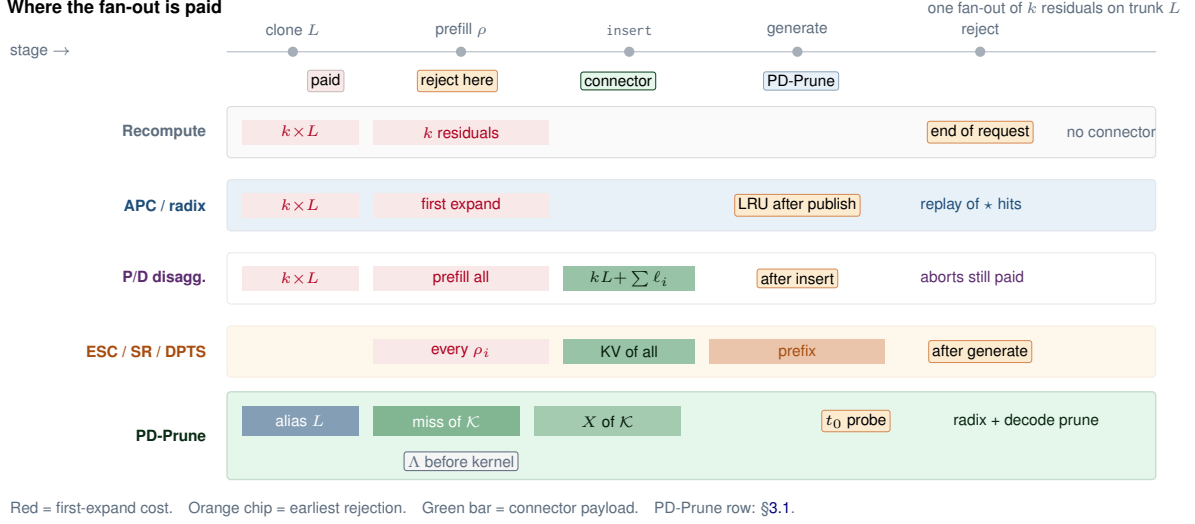


Figure 3: **Where each mechanism acts on one fan-out of k residuals over a trunk of length L .** Red bars denote prefill issued before any sibling is discarded, orange chips the earliest rejection point, and green bars connector volume. Trunk volumes for the prefix-cache rows assume the cold concurrent regime R1; in regimes R2–R4 the trunk is computed at most once (§2.2). The PD-Prune row follows §3.1.

without live descendants is reclaimed up to the nearest ancestor with a live child. Commit publishes the full pages of $x_\star$ to the prefix index so that later requests can hit them. Eviction is node-granular: speculative leaves first, then idle committed leaves, then the spine [16, 33].

B.3 Speculative Prefill of Known Suffixes

A candidate of parent u is a pair $(x^k, \hat{x}^o)$ of a known suffix and a predicted observation, with admission probability p and expected prefix-hit rate q . Known suffixes ($p=q=1$) are scheduled before predicted observations ($q<1$), subject to the idle time $T_{\text{idle}}(u)$, the available KV capacity, and a branch cap. Unlike speculative decoding [15, 6], the object prefilled is a future *prompt* prefix rather than draft output tokens. On commit, the pages covered by $\text{LCP}(x, x_{v^\star})$ become committed, the unmatched tail is prefilled, every sibling is aborted, and jobs tagged with a stale generation are dropped in $O(1)$.

B.4 Admission Details

Scorers. $\text{Score}_{\text{pre}}$ assigns 0.02 to a residual containing a substring of 8–40 characters repeated at least four times, 0.22 to a residual containing malformed mathematical tokens (e.g., runs of asterisks or nan), and otherwise a weighted combination of alphanumeric fraction and character diversity, which is close to 1 for natural prose. $\text{Score}_{\text{dec}}$ applies the same repetition and malformation tests to the generated prefix (0.04 and 0.16), assigns 0.92

once a finished answer marker (#### followed by a number, or `\boxed{}`) appears, and otherwise uses a weighted combination that also rewards digits and operators. Both scorers run on the CPU and do not share parameters.

Ordering of the action. (4) reads decode-side residency first, because a residual already resident on $\mathcal{D}$ costs neither prefill nor transfer. It then protects index 0, so the top-ranked continuation is always served. Only then does it consult the score. A page resident only on $\mathcal{P}$ reduces W_i but not X_i ; the residual is admitted with $W_i = \ell_i - m_i^{\mathcal{P}}$ and $X_i = \ell_i - m_i^{\mathcal{D}}$. Figure 5 illustrates the four actions on one radix tree, and Figure 6 places them on the request path.

Spans. Table 8 summarizes prefill, transfer, and post-abort memory per fan-out for a cold trunk.

B.5 Critical-Path Placement

After abort the live occupancy is M_{spine} , so a pool of M_{pool} bytes holds

$$C = \lfloor M_{\text{pool}}/M \rfloor \quad (12)$$

sessions, with $M=M_{\text{spine}}$ for PD-Prune and $M=M_{\text{APC}}$ while a prefix cache retains aborted residuals. On tick t the chunk budget [2] is split by

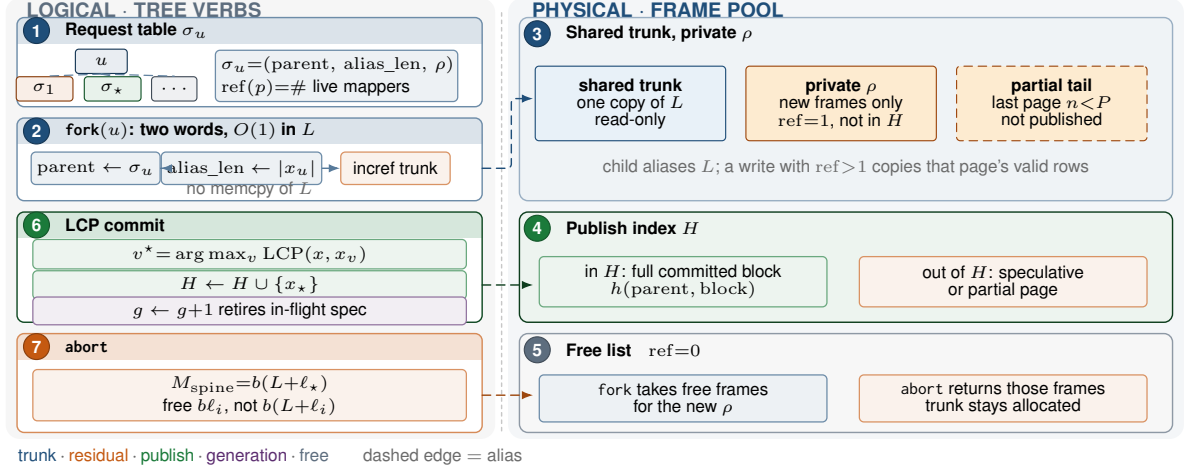


Figure 4: **Context-tree page pool.** fork aliases the trunk in $O(1)$ and allocates only residual pages ①–③. Commit publishes x_* to the index ④⑥. abort returns residual pages only, so the live occupancy is $M_{\text{spine}} = b(L + \ell_*)$ ⑤⑦.

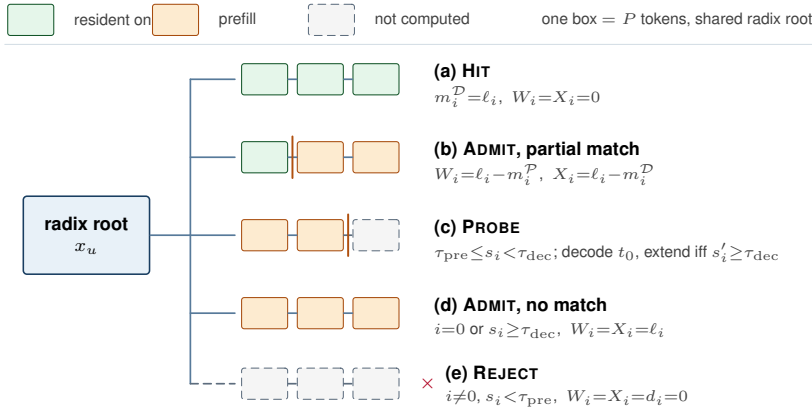


Figure 5: **Four admission actions on one radix tree.** The root is the aliased trunk x_u ; each edge is one page of P tokens. Green pages are resident on $\mathcal{D}$, orange pages are prefilled, and gray pages are never computed. (a) HIT: $m_i^{\mathcal{D}} = \ell_i$, so $W_i = X_i = 0$. (b) ADMIT with a partial match: $W_i = \ell_i - m_i^{\mathcal{P}}, X_i = \ell_i - m_i^{\mathcal{D}}$. (c) PROBE: prefilled and decoded for t_0 tokens, then extended or freed. (d) ADMIT without a match: $W_i = X_i = \ell_i$. (e) REJECT: $W_i = X_i = d_i = 0$.

the committed rate $\underline{s}_r$ [40],

$$B_t^c = \min\left(B_t, \sum_{r \in R_t^c} \max(\underline{s}_r \Delta, 1)\right), \quad (13)$$

$$B_t^s = B_t - B_t^c. \quad (14)$$

B_t^c decodes committed spines; B_t^s prefills admitted residuals or known suffixes ($p=1$) when $T_{\text{pre}}(|w|) \leq T_{\text{idle}}(u)$. Speculatively prefilled tokens are never sampled from.

C Additional Evaluation

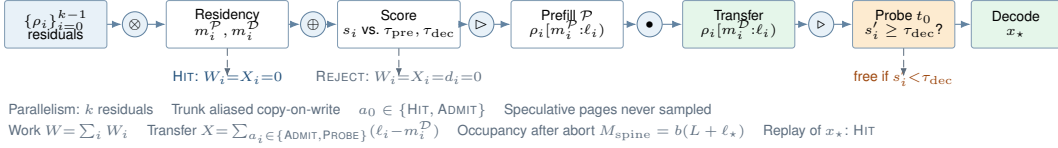
This appendix reports the full GSM8K forest on Qwen3-14B (§C.1), details of the admission and probe experiments (§C.2–§C.5), a Qwen3-8B forest with an answer-marker decode stop (§C.6),

the multi-step sessions (§C.7), cross-model results (§C.8), and sensitivity to width, depth, and trace length (§C.9).

Common setup. All runs use vLLM on four A100-SXM4 40GB GPUs with bfloat16 weights, temperature 0, and physically disaggregated prefill and decode instances connected by a KV connector. Models are Qwen3-14B, Qwen3-8B, Qwen3-4B, Qwen2.5-Math-1.5B, Llama-3-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Llama-8B (R1-8B). Tree-of-Thoughts forests use $k=4$ and $D=512$ (2048 for contest problems) in batches of eight. An answer is correct when its final number matches the reference; unless noted, accuracy is committed-answer accuracy.

ρ_i : i -th residual m_i^P, m_i^D : resident prefix s_i, s'_i : residual and probe score t_0 : probe length W_i : computed prefill X_i : connector tokens a_i : action

(a) PD-Prune admission on a P/D path (Alg. 1)



(b) Cold, concurrent fan-out (regime R1; same k, L, ℓ). Bottleneck in red.

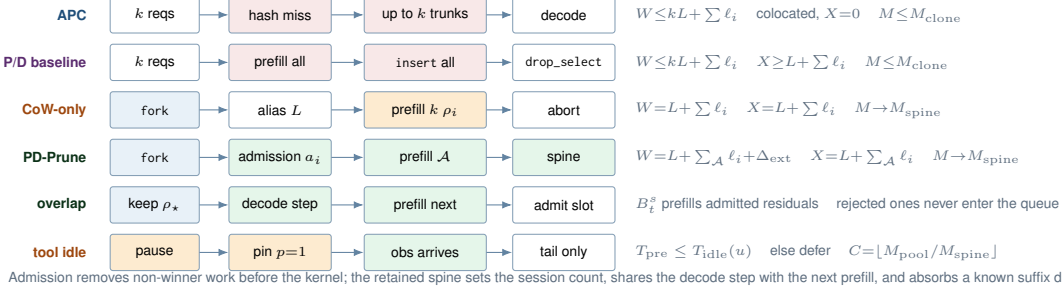


Figure 6: **Admission on the request path.** (a) The action of (4) is evaluated before insert; only admitted and probed residuals reach the prefill kernel. (b) Prefill and connector volume of a cold, concurrent fan-out (regime R1) for APC, a baseline PD deployment, the copy-on-write tree without admission (CoW-only), and PD-Prune. The overlap row prefills the next residual during decode; the tool-interval row prefills a known suffix while $T_{\text{pre}} \leq T_{\text{idle}}$.

Table 8: Per-fan-out spans with a cold trunk ($c_u^P = c_u^D = 0$) and no resident residual pages. Baseline rows assume concurrent admission without deduplication (R1); with serialized admission (R2) their trunk term becomes L . $\mathcal{A}$ is the set of admitted and probed residuals; T_{ab} is the abort cost in pages; “replay” is a later request for the committed spine.

	W	X	M after abort	T_{ab}	trunk copies	replay W
Recompute	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	$\Theta(kL/P)$	k	$L + \ell_*$
APC (R1)	$\leq kL + \sum_i \ell_i$	0	$\leq kL + \sum_i \ell_i$	LRU	$1-k$	0
PD baseline (R1)	$\leq kL + \sum_i \ell_i$	$\geq L + \sum_i \ell_i$	$\leq kL + \sum_i \ell_i$	LRU	$1-k$	0
CoW-only	$L + \sum_i \ell_i$	$L + \sum_i \ell_i$	$L + \ell_*$	$\Theta(\ell_i/P)$	1	0
PD-Prune	$L + \sum_{i \in \mathcal{A}} \ell_i + \Delta_{\text{ext}}$	$L + \sum_{i \in \mathcal{A}} \ell_i$	$L + \ell_*$	$\Theta(\ell_i/P)$	1	0

Table 9: GSM8K test, Qwen3-14B, $\text{tp}=2$, $n=1319$; medians over 165 batches of eight.

	recompute	APC	CoW-only
Committed accuracy	0.847	0.844	0.845
Peak KV, median (tok)	4136	1430	1030
Peak KV, max (tok)	4944	1632	1232
Peak / APC	$2.89 \times$	$1.00 \times$	$0.72 \times$
Fan-out, median (ms)	326	87	140

C.1 GSM8K Forest on Qwen3-14B

The full GSM8K test set ($n=1319$) is run on Qwen3-14B at $\text{tp}=2$ in 165 batches of eight (Table 9). This experiment isolates state management: no residual is rejected, so the copy-on-write tree is compared with recomputation and APC on the same committed path.

Median peak KV is $0.72 \times$ APC and $0.25 \times$ recomputation, and accuracy is within 0.3 points of both. Fan-out latency does not follow the same order: the copy-on-write fan-out takes 140 ms against

87 ms for APC, because fork and abort run on the expand path while an APC hit does not. The same peak ordering ($0.58\text{--}0.80 \times$ APC) holds on seven math datasets for Qwen3-14B, Qwen3-8B, and R1-8B at $\text{tp}=2$ and 4, and peak KV does not change with tensor-parallel width, consistent with (1).

C.2 Probe Admission on MATH-500

The MATH-500 experiments use Qwen3-8B at $\text{tp}=1$, the first 32 problems of level 4 or 5, $D=1024$, and EOS ignored. Half of the k residuals are contest strategies; the other half alternate repetition loops and illegal text. Coverage counts a problem when some branch that ran all 1024 tokens is correct; the committed residual is index 0. Blanket rejection lowers coverage from 12/32 to 6/32 at $k=4$ and from 20/32 to 13/32 at $k=16$, whereas no loop residual yields a correct answer at any k , so rejecting loops does not reduce coverage. At $k=16$ a 128-token probe extends every

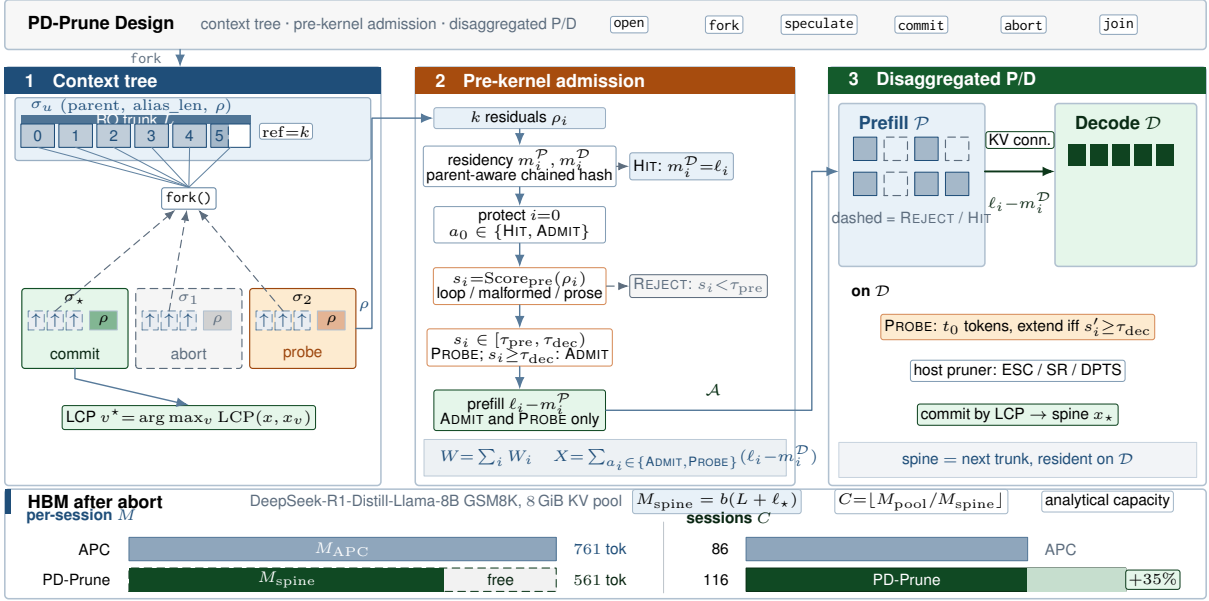


Figure 7: **PD-Prune design**. (1) Context tree: σ_u aliases one trunk, fork is $O(1)$, and (11) commits the winner. (2) Pre-kernel admission assigns one action per residual in the order of (4); only $\mathcal{A}$ (ADMIT and PROBE) reaches the prefill kernel. (3) Disaggregated execution: the connector carries only decoder-missing pages, $\mathcal{D}$ probes and extends uncertain residuals under the host pruner, and the committed spine becomes the next trunk. Bottom: HBM occupancy after abort and the resulting session capacity.

illegal residual (128/128) and restores 20/32 in 72.2 s instead of 104.1 s; probing the loops as well extends none of them and costs 74.4 s. At $k=8$, $t_0=64$ reaches 16/32 in 42.0 s, above the full fan-out (15/32 in 52.5 s); at $k=4$ no probe recovers 12/32, and reject-only admission (11/32 in 23.0 s) is the closest setting. Figure 9 plots Table 4.

C.3 Admission on GSM8K

The GSM8K admission experiments use the branching mix with $n=16$ and $D=512$ on Qwen3-8B at $\text{tp}=1$. Because the heuristic scores take three values on this mix (loop 0.02, illegal 0.22, live ≈ 1), any $\tau_{\text{pre}} \in (0.02, 0.22]$ yields the same admission set as 0.15, and any $\tau_{\text{pre}} \in (0.22, 1)$ yields the same set as 0.45; a denser threshold sweep therefore adds no information on this workload. Figure 10 plots ESC across widths and the three hosts at $k=16$. At $k=8$, reject-only admission keeps ESC at 16/16 coverage (10.84 to 10.19 s) and raises Speculative Rejection and DPTS from 15/16 to 16/16.

C.4 Decode-Only, Prefill-Only, and Combined Pruning

Table 16 (Appendix D.3) factorizes the $k=16$ results. Decode-only pruning leaves W at 5008 tokens; prefill-only admission decodes every admitted residual to D ; the combination retains the prefill

reduction and adds the host’s decode-time termination. At $\tau_{\text{pre}}=0.45$ the hosts terminate no additional residuals, because every residual they would terminate has already been rejected.

C.5 Score-Based versus Winner-Only Admission

Admitting only index 0 is a different policy from scoring every residual. At $k=16$ and $\tau=0.45$, winner-only admission prefills 414 tokens but lowers coverage to 11/16 under ESC and 9/16 under Speculative Rejection, while score-based admission of the eight residuals above τ prefills 2768 tokens and retains 16/16 (Table 17). Committed accuracy is similar for both policies (9–11 of 16), so the difference lies in the coverage available to a better selector.

Capacity from measured peaks. For R1-8B on GSM8K ($n=16$, $k=4$), measured median peaks are 761 tokens under APC and 561 under the context tree. With 32 layers, 8 KV heads of dimension 128, and bfloat16 , one token occupies 131072 bytes, so an 8 GiB KV budget holds 65536 tokens, i.e., 86 APC trees or 116 spines by (12) (+35%). This is an analytical capacity bound; we do not report SLO-constrained throughput, which would additionally depend on scheduling and batching.

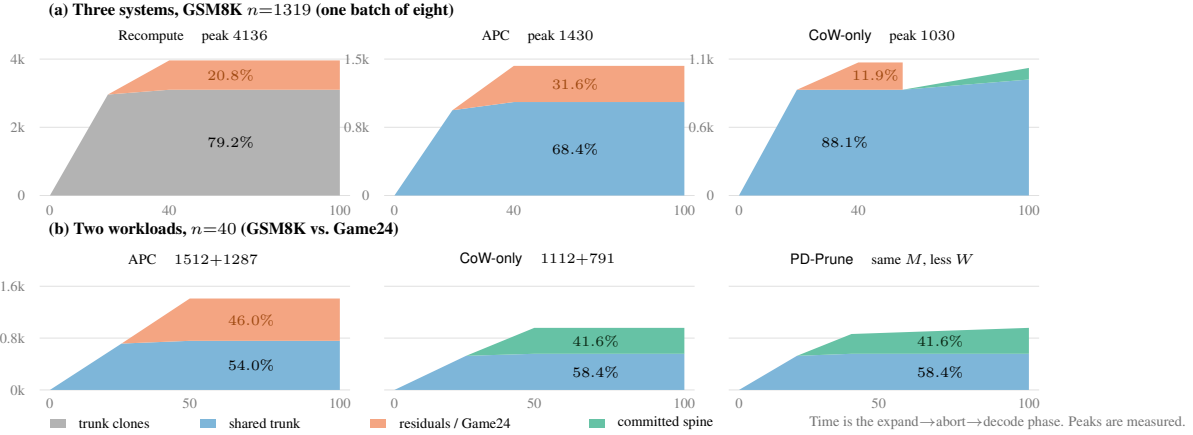


Figure 8: **Stacked KV occupancy** over expand, abort, and decode. (a) One GSM8K batch of eight on Qwen3-14B (per-panel y -scale; peaks 4136, 1430, 1030). (b) GSM8K (bottom) and Game of 24 (top), $n=40$, for APC, CoW-only, and PD-Prune; CoW-only and PD-Prune share M , and PD-Prune computes fewer residual tokens.

MATH-500, Qwen3-8B, $n=32$, $D=1024$, $\tau=0.45$

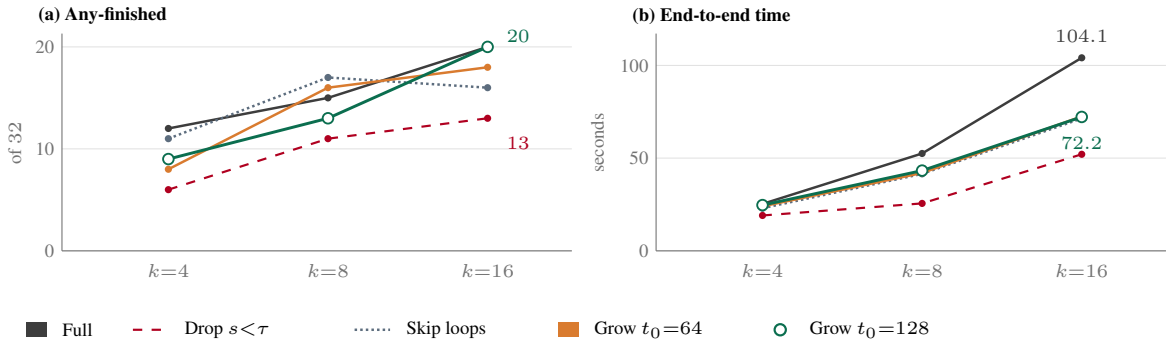


Figure 9: **Probe admission on MATH-500** (Table 4). Blanket rejection is fast but loses coverage; a $t_0=128$ probe returns to 20/32 at $k=16$ in 72.2 s instead of 104.1 s.

C.6 Qwen3-8B Forest with an Answer-Marker Stop

This experiment runs Qwen3-8B on GSM8K with $n=80$ and $D=512$ at $tp=2$ and 4; Game of 24 uses Tree-of-Thoughts with $k=4$, and HumanEval uses a ReAct wrap-and-recover loop with $k=2$. The PD-Prune runs additionally stop committed decode when the last non-empty line contains a finished answer, which reduces D^+ to about one third. On this trace the scorer admits every issued residual, so the end-to-end ratio in Table 10 reflects the decode stop rather than admission.

Peak KV is $0.75\times$ APC on GSM8K, $0.62\times$ on Game of 24, and $0.96\times$ on HumanEval, where $k=2$ makes the residual a small fraction of the context. Accuracy is 1–3 points below APC (0.863 vs. 0.900 at $tp=2$; 0.925 vs. 0.938 at $tp=4$), which we attribute to the decode stop truncating a few answers.

Table 10: Qwen3-8B forest, $n=80$, $D=512$. PD-Prune includes an answer-marker decode stop. Peak KV does not depend on tp ; G24 and HE are the Game of 24 and HumanEval peaks.

	peak	D^+	e2e (s)	acc.	e2e/APC	G24	HE
APC, $tp=2$	1582	40960	36.0	0.900	—	1289	1970
PD-Prune, $tp=2$	1182	13356	28.2	0.863	0.78	793	1882
APC, $tp=4$	1582	40960	22.7	0.938	—	1289	1970
PD-Prune, $tp=4$	1182	12826	16.9	0.925	0.75	793	1882

C.7 Multi-Step Tree-of-Thoughts

Each problem is one session of $k=4$ candidates over three steps on Qwen3-4B and Qwen3-8B ($tp=2$, $n=16$), issued as its own request. Steps 1 and 2 decode a 32-token step, abort the non-selected candidates, and promote the selected spine to the next trunk; step 3 decodes the final answer with $D=512$ (GSM8K) or $D=256$ (Game of 24). On Game of 24 the steps propose an equation, continue on the remaining cards, and write the final ex-

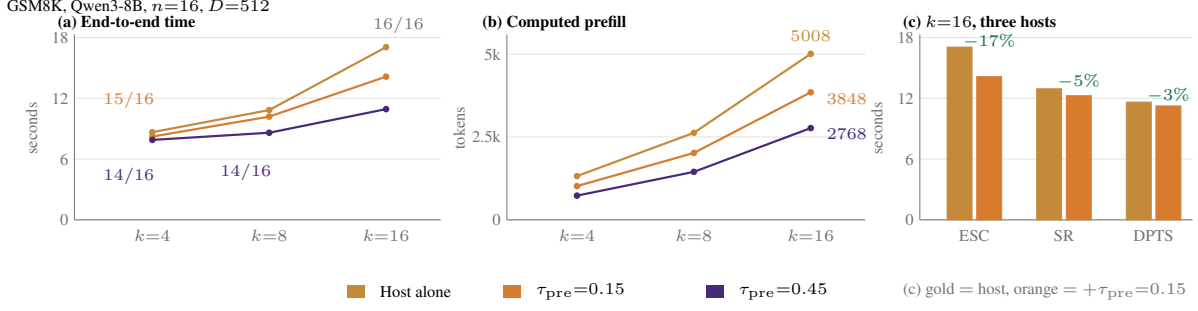


Figure 10: **Admission threshold on GSM8K** (Tables 2 and 3). (a) ESC wall-clock across k , labeled with any-finished coverage. (b) Computed prefill: -23% at $\tau_{\text{pre}}=0.15$ and -45% at 0.45 . (c) Reject-only admission in front of each host at $k=16$.

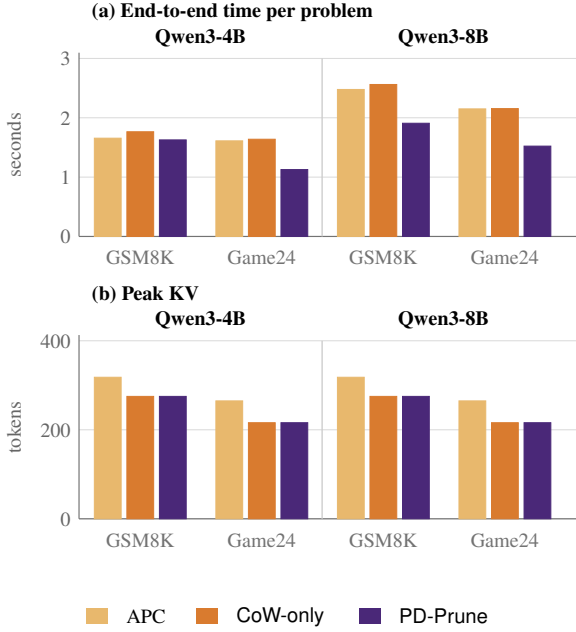


Figure 11: **Three-step Tree-of-Thoughts** ($n=16$, $\text{tp}=2$). (a) End-to-end seconds per problem; PD-Prune includes an answer-marker decode stop. (b) Peak KV of the live spine.

pression; on GSM8K they set up the algebra, revise the previous step, and state the final number. Admission runs before every fan-out, and the 32-token step is decoded only for admitted residuals. Per-phase timings are not comparable across systems, because the APC harness attributes step decoding to the expansion phase whereas ours attributes it to decode; Table 6 therefore reports end-to-end time only.

C.8 Cross-Model Results

The forest is repeated on six families ($n=40$, $D=512$, $\text{tp}=2$ and 4). APC retains the shared trunk and all residuals, $M_{\text{CoW}} = b(L + \sum_i \ell_i)$,

Table 11: Spine peak and an optional answer-marker decode stop ($n=32$, $\text{tp}=2$; GSM8K $D=512$, Game of 24 $D=256$). The PD-Prune column applies both the context tree and the stop.

Model	Metric	APC	PD-Prune
R1-8B	GSM8K peak	1493	1093
	GSM8K acc.	0.500	0.562
	GSM8K D^+	16384	12693
	Game24 peak	1064	616
Llama-3-8B	GSM8K peak	1485	1085
	GSM8K acc.	0.719	0.688
	GSM8K D^+	16384	5101
	Game24 peak	1224	736
Mistral-7B	GSM8K peak	1521	1121
	GSM8K acc.	0.344	0.188
	GSM8K D^+	16384	6723
	Game24 peak	1296	728

while the context tree retains $M_{\text{spine}} = b(L + \ell_*)$; their ratio $(L + \ell_*) / (L + \sum_i \ell_i)$ is independent of the model and of tensor-parallel width. For similar residual lengths it reduces to $(L + \ell) / (L + k\ell)$, which decreases as L/ℓ decreases: 0.72 – 0.75 on GSM8K with a long trunk and 0.57 – 0.61 on Game of 24 with a short one (Figure 12). Relative to recomputation, which stores M_{clone} , the spine is $0.25\times$.

Figure 13 repeats the comparison at $n=80$; Figure 14 shows per-item decode lengths.

Table 11 isolates the answer-marker stop on three further models ($n=32$). The stop reduces D^+ by 23 – 69% and leaves the spine peak unchanged, but its effect on accuracy is model-dependent: $+2$ problems on R1-8B, -1 on Llama-3-8B, and -5 on Mistral-7B (0.344 to 0.188), where the model emits intermediate answer markers before finishing. The stop is therefore not a safe default and is not part of the admission results in §4.

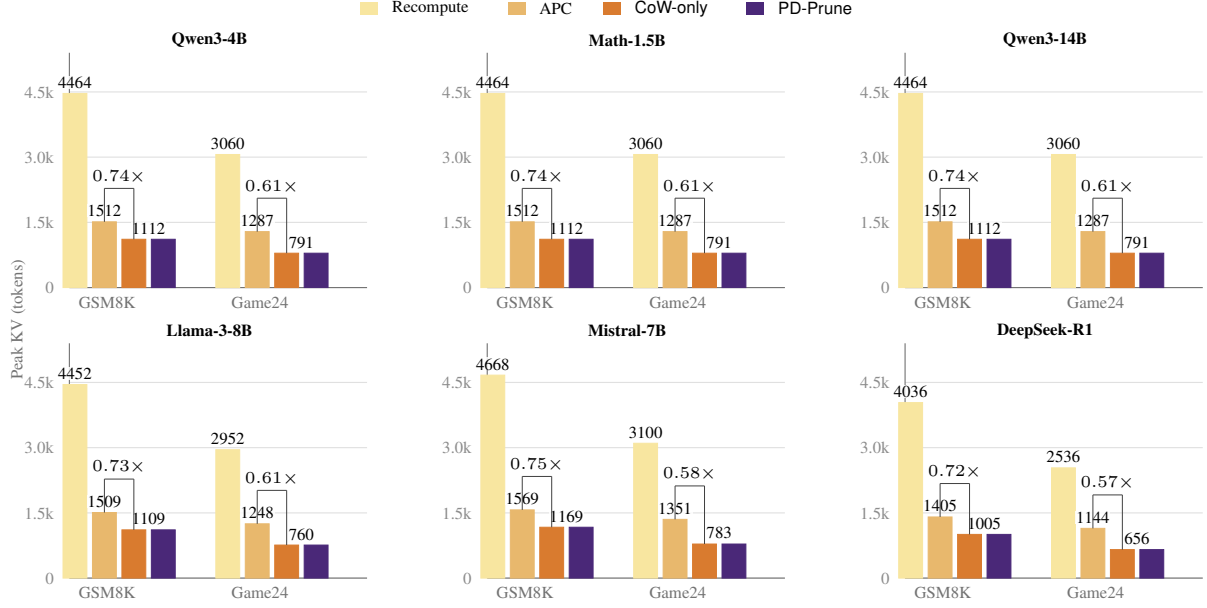


Figure 12: **Peak KV across six model families** ($n=40$, $tp=2$). Bars are recompute, APC, CoW-only, and PD-Prune; CoW-only and PD-Prune retain the same spine. The annotated ratio is $M_{\text{spine}}/M_{\text{APC}}$.

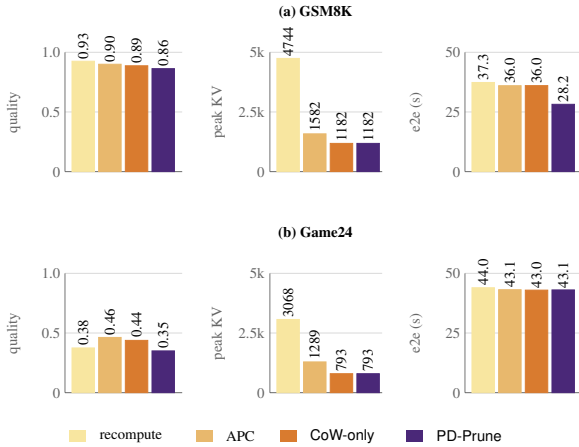


Figure 13: **Four systems on Qwen3-8B** ($n=80$, $tp=2$, $D=512$). Each panel has its own axis; for peak KV and end-to-end time shorter is better. (a) GSM8K. (b) Game of 24. CoW-only and PD-Prune lower peak KV to $0.75\times$ (GSM8K) and $0.62\times$ (Game of 24) APC; the end-to-end reduction of PD-Prune on GSM8K (36.0 to 28.2 s) is due to the answer-marker stop.

C.9 Width, Depth, and Long Traces

GSM8K runs use $n=16$, $D=512$, and $tp=2$ on Llama-3-8B, Mistral-7B, and R1-8B, varying k or adding a second Tree-of-Thoughts level (Table 12 and Figure 15). The spine peak is invariant in k , while the APC peak grows with the union of residuals, so the ratio falls from 0.89–0.90 at $k=2$ to 0.53–0.55 at $k=8$; at depth 2 it is 0.75–0.76. Admission leaves the peak unchanged; at $k=8$ it re-

Table 12: GSM8K peak KV (tokens) versus branching ($n=16$, $tp=2$, $D=512$). PD-Prune has the same peak as CoW-only.

Model		$k=2$	$k=4$	$k=8$	depth 2
Llama-3-8B	APC	610	750	998	814
	CoW-only	550	550	550	614
	ratio	0.90	0.73	0.55	0.75
Mistral-7B	APC	629	769	1033	841
	CoW-only	569	569	569	641
	ratio	0.90	0.74	0.55	0.76
R1-8B	APC	629	761	1053	841
	CoW-only	561	561	561	641
	ratio	0.89	0.74	0.53	0.76

jects 28 branches and reduces fan-out time relative to CoW-only from 307 to 179 ms on Llama-3-8B (APC: 224 ms), 349 to 191 ms on Mistral-7B, and 363 to 194 ms on R1-8B. With a batch of 16, the ratio is 0.72–0.73.

Table 13 reports contest problems with $D=2048$ (MATH-500 $n=200$, AIME $n=73$). The peak ratio is 0.78–0.91, and accuracy is within 3 points of APC on MATH-500 and within one problem on AIME. We do not compare end-to-end time here: the APC harness decodes the full budget for every item, whereas ours stops at EOS, so time differences would reflect the harnesses rather than the methods. The trunk token counts of the two harnesses also differ slightly (e.g., 95 vs. 73 on R1-8B MATH-500) because of prompt templating, which adds an offset to the ratios.

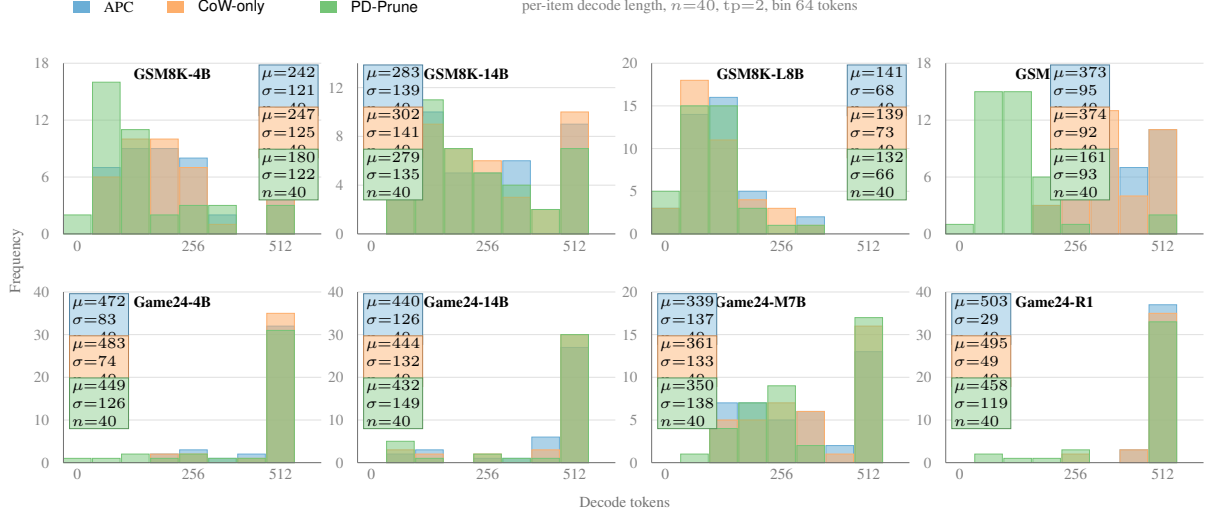


Figure 14: **Per-item decode-length histograms** ($n=40$, $tp=2$, $D=512$, bin width 64). GSM8K (top) and Game of 24 (bottom) overlay APC, CoW-only, and PD-Prune; boxes report μ , σ , and n . Llama-3-8B on Game of 24 is omitted because every output has length 1; Mistral-7B is shown instead.

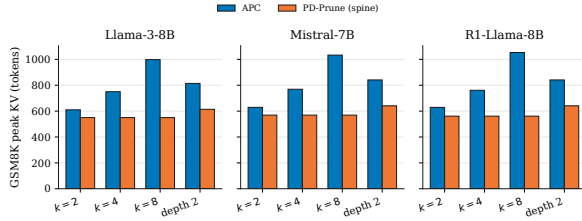


Figure 15: **GSM8K peak KV versus branching** ($n=16$, $tp=2$, $D=512$). The spine peak (PD-Prune and CoW-only) is invariant in k , while the APC peak grows with k .

D Full Ablation Tables

This appendix gives the full ablations behind §4. All runs use Qwen3-8B ($tp=1$, temperature 0, prefix caching) on the branching mix of §4: GSM8K with $n=16$, $D=512$, and MATH-500 (first 32 problems of level ≥ 4) with $D=1024$. EOS is ignored, so decode volume is determined by the policy. We report computed prefill W (prompt minus cache hits), decode tokens D^+ , wall-clock T_{e2e} in seconds (trunk warm-up plus fan-out), committed accuracy (comm.), and any-finished coverage (any).

Notation for this appendix.

Table 13: Contest problems, $D=2048$, $k=4$, $tp=2$. All rows of a model come from the same run.

Setting	System	peak	/ APC	acc.
R1-8B MATH	APC	1863	—	0.560
	CoW-only	1599	0.86	0.530
	PD-Prune	1599	0.86	0.530
R1-8B AIME	APC	2118	—	0.041
	CoW-only	1662	0.78	0.041
	PD-Prune	1662	0.78	0.041
Qwen3-4B MATH	APC	1937	—	0.620
	CoW-only	1770	0.91	0.625
	PD-Prune	1770	0.91	0.600
Qwen3-4B AIME	APC	2150	—	0.027
	CoW-only	1794	0.83	0.041
	PD-Prune	1794	0.83	0.041

Symbol	Domain	Meaning
k	$\{4, 8, 16\}$	fan-out width
τ_{pre}	$\{0, 0.10, 0.15, 0.30, 0.45\}$	rejection threshold on $\text{Score}_{pre}(\rho_i)$
τ_{dec}	0.45	extension threshold on $\text{Score}_{dec}(y_i)$
t_0	$\{64, 128\}$	probe length (MATH-500)
admit	$\mathbb{N}$	residuals that are admitted or probed
W	tokens	computed prefill
D^+	tokens	decode tokens issued
T_{e2e}	s	end-to-end wall-clock
host	$\{\text{ESC}, \text{SR}, \text{DPTS}\}$	decode-time pruner

D.1 Host $\times$ $\tau_{pre} \times k$ (GSM8K)

Table 14 is the complete host $\times$ threshold $\times$ width matrix. All base hosts at fixed k share the same W , since hosts act after prefill (§3.3). Rejecting loops ($\tau_{pre}=0.15$) removes a quarter of the residuals and keeps coverage at or above the host baseline except ESC at $k=4$ (−1). Blanket rejection at 0.45 removes half of the residuals and is faster, but lowers coverage at $k=4$ and $k=8$ for ESC.

Table 14: Full host ablation on GSM8K (Qwen3-8B, branching mix, $n=16$, $D=512$). Each cell: any / comm. / T_{e2e} s / W / admit. Base rows have no admission; other rows reject $s_i < \tau_{pre}$ and admit the rest.

Host	Gate	$k=4$	$k=8$	$k=16$
ESC	base	16/16, 9/16, 8.65, 1320, 4	16/16, 11/16, 10.84, 2624, 8	16/16, 11/16, 17.05, 5008, 16
ESC	$\tau_{pre}=0.15$	15/16, 9/16, 8.22, 1018, 3	16/16, 9/16, 10.19, 2020, 6	16/16, 11/16, 14.13, 3848, 12
ESC	$\tau_{pre}=0.45$	14/16, 11/16, 7.89, 732, 2	14/16, 9/16, 8.60, 1448, 4	16/16, 10/16, 10.92, 2768, 8
SR	base	13/16, 8/16, 8.19, 1320, 4	15/16, 9/16, 9.61, 2624, 8	16/16, 12/16, 13.14, 5008, 16
SR	$\tau_{pre}=0.15$	13/16, 9/16, 8.03, 1018, 3	16/16, 9/16, 9.37, 2020, 6	16/16, 10/16, 12.47, 3848, 12
SR	$\tau_{pre}=0.45$	12/16, 9/16, 7.89, 732, 2	15/16, 9/16, 8.64, 1448, 4	16/16, 10/16, 10.80, 2768, 8
DPTS	base	13/16, 8/16, 8.00, 1320, 4	15/16, 9/16, 8.97, 2624, 8	16/16, 12/16, 11.86, 5008, 16
DPTS	$\tau_{pre}=0.15$	13/16, 9/16, 7.93, 1018, 3	16/16, 8/16, 8.95, 2020, 6	16/16, 12/16, 11.49, 3848, 12
DPTS	$\tau_{pre}=0.45$	13/16, 9/16, 7.88, 732, 2	15/16, 9/16, 8.67, 1448, 4	16/16, 10/16, 10.90, 2768, 8

Table 15: Prefill-only admission (no host cut; every admitted residual decodes to D).

k	τ_{pre}	admit	W	D^+	T_{e2e}	any	comm.
4	0.00	4	1320	32768	8.65	16/16	9/16
4	0.10	3	1018	24576	8.20	15/16	9/16
4	0.30	2	732	16384	7.89	14/16	11/16
8	0.00	8	2624	65536	10.79	16/16	9/16
8	0.10	6	2020	49152	10.21	16/16	10/16
8	0.30	4	1448	32768	8.64	15/16	9/16
16	0.00	16	5008	131072	16.99	16/16	8/16
16	0.10	12	3848	98304	14.14	16/16	9/16
16	0.30	8	2768	65536	10.91	16/16	10/16

Analysis. At every k , the three base hosts report identical W and admit counts. Admission changes only that column: $3k/4$ residuals are admitted at $\tau_{pre}=0.15$ (live and illegal; loops rejected) and $k/2$ at 0.45 (live only). Relative T_{e2e} cuts under ESC+0.15 grow with k : $(8.65-8.22)/8.65=5\%$, $(10.84-10.19)/10.84=6\%$, $(17.05-14.13)/17.05=17\%$. Because SR and DPTS already terminate loops during decoding, the same threshold yields smaller relative reductions (5% and 3% at $k=16$).

D.2 Prefill-only ESC bars

Table 15 isolates the prefill-side score: no host criterion is applied, and every admitted residual is decoded to D . Since Score_{pre} takes three values on the mix (loop 0.02, illegal 0.22, live ≈ 1), the thresholds 0.00, 0.10, and 0.30 yield the three distinct admission sets; 0.10 is equivalent to 0.15, and 0.30 to 0.45. At $k=16$, the $\tau_{pre}=0$ row coincides in configuration with the ESC baseline of Table 14, because the ESC stopping criterion is not triggered on this mix. The two runs nevertheless commit 8/16 and 11/16 correct answers, which we take as an estimate of the run-to-run variance of committed accuracy at $n=16$.

D.3 Decode-only, prefill-only, and combined ($k=16$)

Table 16 factorizes the $k=16$ results. “Stopped” denotes the number of admitted residuals terminated by the host’s decode-time criterion. Combined rows apply admission first and then the host criterion to the admitted set.

D.4 Threshold-Based versus Winner-Only Admission

Two policies share $\tau=0.45$ but differ in how many residuals are admitted. *Score* admits every i with $s_i \geq \tau$ and index 0 ($k/2$ residuals on the mix). *Winner-only* admits index 0 alone ($W=414$). Table 17 shows that winner-only lowers coverage (ESC 11/16, Speculative Rejection 9/16 at $k=16$), while score-based admission keeps 16/16.

D.5 MATH-500: Blanket Rejection, Reject-Only Admission, and Probing

Table 18 compares the full fan-out, blanket rejection, reject-only admission, and truncating loop residuals after 128 decode tokens. Table 19 probes the band $[\tau_{pre}, \tau_{dec})$ for t_0 tokens and extends to D iff $s'_i \geq \tau_{dec}=0.45$.

Summary. (i) Blanket rejection of every $s_i < \tau_{dec}$ loses coverage: 20/32 falls to 13/32 at $k=16$. (ii) Rejecting only loops costs at most one problem at $k \leq 8$ but recovers only 16/32 at $k=16$. (iii) A $t_0=128$ probe restores 20/32 at $k=16$ and cuts 104.10 s to 72.21 s (-31%); at $k=8$ the shorter probe $t_0=64$ is the accuracy-preserving point (16/32 in 42.01 s). (iv) Probing loop residuals as well extends none of them and increases wall-clock time (74.44 s at $k=16$). (v) At $k=16$, probe extension recomputes evicted pages, so W exceeds that of the full fan-out although D^+ and wall-clock time decrease; retaining probe pages until the extension decision would eliminate this

Table 16: Factorization at $k=16$ (GSM8K, Qwen3-8B). Decode-only: τ_{pre} absent, $W=5008$. Prefill-only: host cut disabled, decode to D . Combined: both. Any-finished is 16/16 on every row. Prefill-only and combined rows come from separate runs.

Factor	Setting	admit	stopped	W	D^+	T_{e2e}	comm.
<i>Decode-only</i>							
	ESC ($D=512$)	16	0	5008	131072	17.05	11/16
	SR ($\tau=256$)	16	8	5008	98304	13.14	12/16
	DPTS ($d=100$)	16	8	5008	78336	11.86	12/16
<i>Prefill-only</i>							
	$\tau_{\text{pre}}=0.15$	12	0	3848	98304	14.14	11/16
	$\tau_{\text{pre}}=0.45$	8	0	2768	65536	10.91	10/16
<i>Combined</i>							
	ESC+0.15	12	0	3848	98304	14.13	11/16
	SR+0.15	12	4	3848	81920	12.47	10/16
	DPTS+0.15	12	4	3848	71936	11.49	12/16
	ESC+0.45	8	0	2768	65536	10.92	10/16
	SR+0.45	8	0	2768	65536	10.80	10/16
	DPTS+0.45	8	0	2768	65536	10.90	10/16

Table 17: Score-based versus winner-only admission at $\tau=0.45$ (GSM8K, Qwen3-8B). Rows come from a separate run from Table 14.

k	Policy	Host	admit	W	T_{e2e}	any	comm.
16	base	ESC	16	5008	17.02	16/16	10/16
16	score	ESC	8	2768	10.91	16/16	10/16
16	winner-only	ESC	1	414	7.30	11/16	11/16
16	base	SR	16	5008	13.12	16/16	12/16
16	score	SR	8	2768	10.84	16/16	10/16
16	winner-only	SR	1	414	7.29	9/16	9/16
16	base	DPTS	16	5008	11.85	16/16	10/16
16	score	DPTS	8	2768	10.89	16/16	9/16
16	winner-only	DPTS	1	414	7.30	11/16	11/16
8	winner-only	ESC	1	414	7.28	11/16	11/16
8	score	ESC	4	1448	8.65	14/16	9/16
4	winner-only	ESC	1	414	7.29	10/16	10/16
4	score	ESC	2	732	7.88	14/16	11/16

term.

D.6 Token Accounting Checks

Across Tables 14 and 15, the measured prefill satisfies the predictions of Proposition 1 for a warm trunk: the three hosts report identical W at every k , and admission removes exactly the computed tokens of the rejected residuals. Because residual lengths differ across classes, the reductions are not fixed fractions of W :

	$k=4$	$k=8$	$k=16$
W , no admission	1320	2624	5008
W , $\tau_{\text{pre}}=0.15$	1018 (0.77)	2020 (0.77)	3848 (0.77)
W , $\tau_{\text{pre}}=0.45$	732 (0.55)	1448 (0.55)	2768 (0.55)

Loops therefore account for about 23% of residual prefill and illegal residuals for a further 22%, although each class is a quarter of the residuals. Combined rows never exceed the prefill-only W at the same τ_{pre} .

E Extended Related Work

Inference-time search. Chain-of-thought prompting elicits intermediate steps [32]; Tree-of-Thoughts and Monte Carlo tree search expand,

score, and backtrack over such steps [35, 39]; self-consistency samples complete traces and takes a majority [31]. Accuracy and token cost both grow with the number of traces. EquivPruner observes that textually distinct steps can be the same mathematical action and retains one representative per equivalence class [21]. PD-Prune neither changes the search procedure nor decides semantic equivalence; it decides, for the residuals a harness proposes, which are prefilled, transferred, and decoded.

Path pruning. ESC stops sampling once a window agrees [17]; Speculative Rejection halts low-reward partial generations at a horizon [26]; DPTS early-stops children after a mini-step [10]; STOP categorizes pruning signals as internal or external and learned or not, and trains an internal futility token [4]. These methods read generated tokens or compare written steps, so they act on the decode side of a PD deployment. PD-Prune adds a pre-kernel decision and a decode-side probe around them and leaves their decode-time cut unchanged (§C.4); a learned signal such as STOP’s could re-

Table 18: MATH-500 level ≥ 4 , Qwen3-8B, $n=32$, $D=1024$. “Blanket” rejects every $s_i < 0.45$. “Reject-only” rejects loops only. “Decode loops@128” admits every residual but limits the decoding of loops to 128 tokens.

k	Setting	admit	W	D^+	T_{e2e}	any
4	full	4	2844	131072	25.34	12/32
4	blanket	2	1566	65536	19.11	6/32
4	reject-only	3	2189	98304	23.05	11/32
4	short probe@128	4	2844	102400	23.14	10/32
8	full	8	5592	262144	52.50	15/32
8	blanket	4	3036	131072	25.60	11/32
8	reject-only	6	4282	196608	41.53	17/32
8	short probe@128	8	5592	204800	42.26	15/32
16	full	16	12320	524288	104.10	20/32
16	blanket	8	6200	262144	52.05	13/32
16	reject-only	12	9332	393216	71.39	16/32
16	short probe@128	16	11856	409600	74.03	17/32

Table 19: Probe admission on MATH-500. Probe@ t_0 : loops rejected; illegal residuals probed and extended iff $s'_i \geq 0.45$. Probe-loops: loops are probed as well (none is extended). At $k=16$, W includes the recomputation Δ_{ext} of evicted probe pages; it is about 19.4k tokens at $t_0=128$ and 11.2k at $t_0=64$ relative to reject-only (9332).

k	Setting	admit	W	D^+	T_{e2e}	any
4	probe@64	3	2460	98304	23.96	8/32
4	probe@128	3	2460	98304	24.70	9/32
4	probe-loops@128	4	3115	102400	24.89	10/32
8	probe@64	6	4824	196608	42.01	16/32
8	probe@128	6	4824	196608	43.30	13/32
8	probe-loops@128	8	6134	204800	43.32	15/32
16	probe@64	12	20544	393216	72.50	18/32
16	probe@128	12	28736	393216	72.21	20/32
16	probe-loops@128	16	31260	409600	74.44	20/32

place $\text{Score}_{\text{dec}}$.

Request-level serving. Orca introduced iteration-level scheduling [37]; vLLM pages the KV cache and hashes full blocks for prefix caching [14]; SGLang shares prefixes through a radix tree [41]; SARATHI and Sarathi-Serve chunk prefills to co-schedule them with decode [3, 2]; DistServe and Splitwise disaggregate prefill and decode [42, 25]. Whether these engines share a trunk across concurrent siblings depends on the cache regime (§2.2); in all cases they serve every request they receive. PD-Prune uses their residency information as an input to admission and their connector as the channel whose volume admission reduces.

Program-, workflow-, and conversation-level scheduling. Agentix schedules by program-level attained service [23]; JITServe allocates just enough bandwidth under heterogeneous SLOs [40]; Hermes, ThunderAgent, and SAGA schedule at program or workflow granularity [22, 13, 11]; Parrot, Ayo, and Pythia exploit application structure [20, 27, 38]. ConServe disaggregates prefill and decode scheduling at conversation granularity based on observed agent behavior [9]. These systems decide when and where requests run; PD-

Prune decides which sibling continuations become requests and at which stage a discarded one stops consuming resources.

KV retention, overlap, and copy-on-write. InferCept, Continuum, and MORI preserve or place context across tool calls [1, 16, 33]. Sutradhara overlaps tool execution with the prefill of a single tool-independent prefix [5]; SmoothAgent transforms long-horizon context; IdleSpec uses idle time for speculative planning to improve accuracy rather than KV reuse [24, 7]. ForkKV shares a base KV cache copy-on-write across multi-LoRA agents [29], which is closest to our context tree; PD-Prune applies copy-on-write to sibling continuations of one agent step and couples it with admission and spine retention. CachePrune edits KV neurons so that a model ignores instructions embedded in retrieved context [28]; it changes attention state within one request and does not decide which siblings are served.

Speculative decoding. Speculative decoding and multi-head drafters propose *output* tokens and verify them against the target model [15, 6, 18]. The object PD-Prune may prefill speculatively is a future *prompt* prefix, such as a known tool-call suffix

(Appendix B.3); speculatively prefilled pages are never sampled from, and (11) binds the prompt that actually arrives.